



IslamicLegalBench: Evaluating LLMs knowledge and reasoning of islamic law across 1,200 years of Islamic pluralist legal traditions

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Abstract

As millions of Muslims worldwide turn to LLMs like GPT, Claude, and DeepSeek for religious guidance, a critical question emerges: *Can these AI systems reliably reason about Islamic law?* This paper introduces IslamicLegalBench, the first multi-school benchmark for evaluating LLM performance across a broad range of Islamic jurisprudence (*fiqh*). Drawing from 37 foundational legal texts spanning 1,200 years and seven schools of jurisprudence, we create 718 evaluation instances across 13 tasks, manually collected and organized by complexity, from basic recall to sophisticated reasoning including legal rationale identification (*'illah*), analogical application (*qiyās*), and cross-school synthesis. Our evaluation of nine state-of-the-art LLMs reveals significant limitations. Even the best model achieves only 67.65% correctness with 21.25% hallucination; several models achieved correctness below 35% and hallucination exceeding 55%. Furthermore, few-shot prompting yields no consistent benefit; per-model changes range from -3.64 to +4.16 points, with gains concentrated in already-weak models, *consistent with insufficient Islamic legal knowledge in current training data that prompting, under closed-book conditions, could not compensate for*. Our analysis reveals why: moderate-complexity tasks (from a human expert's perspective) requiring exact, verbatim knowledge (e.g., enumerating contract conditions, synthesizing statutory articles) show consistently high error rates and hallucination rates up to 73%, whereas high-complexity tasks show better performance because models rely on semantic generalization and verbose reasoning, projecting competence while lacking precise textual understanding. False premise detection reveals risky sycophantic behavior: under few-shot prompting, 5 of 9 models accept misleading assumptions at rates exceeding 40% (worst: 86.27%). Critically, few-shot prompting worsens sycophancy by 3.49 percentage points. The strong and statistically significant negative correlation (Pearson's $r =$

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−0.91, $n = 9$, $p < 0.01$) between the false premise acceptance rate and overall performance indicates that models with higher false Islamic query acceptance rates tend to exhibit lower overall reasoning accuracy. Our findings carry important implications: the path forward for Islamic NLP lies not in lightweight post-hoc techniques such as prompt or instruction tuning but in enriching knowledge. The Islamic NLP community must prioritize training models on comprehensive, large-scale Islamic legal corpora that span classical Hadith collections, jurisprudential works from the major Islamic law schools (*madhabs*), and codified legal compendia. *IslamicLegalBench* provides the first systematic evaluation framework for Islamic legal AI systems, revealing critical limitations in platforms that Muslims increasingly rely on for spiritual guidance.

Keywords LLM Benchmarking · Islamic NLP · Hallucination · GenAI · Islamic Law

1 Introduction

As large language models (LLMs) are increasingly deployed in sensitive domains such as law, medicine, and religion, the challenge of evaluating their reliability grows ever more pressing. Islamic law (*fiqh*) is especially significant in this regard: it governs the ethical norms, legal rulings, and everyday practices of nearly two billion Muslims worldwide, shaping areas as diverse as contracts, finance, family relations, criminal behavior, dietary norms, among other forms of behavioral governance (Khair and Sawalha 2025). Islamic jurisprudence draws upon a complex interplay of sources, such as the Qur'an, Prophetic traditions (*Sunnah*), scholarly consensus (*ijmā'*), and analogical reasoning (*qiyās*), and is further marked by a plurality of interpretive traditions. The four major Sunni schools and various Shi'i traditions each employ distinct methodological frameworks (*uṣūl al-fiqh*) and develop their own interpretive approaches. Unlike codified Western legal systems that rely heavily on precedent or codified legislation, Islamic jurisprudence emphasizes deep engagement with foundational texts, nuanced interpretive reasoning, and sensitivity to social and historical context (Atif et al. 2025). This combination of doctrinal diversity, methodological rigor, and context-dependent application makes *fiqh* a technically demanding and ethically consequential domain for AI evaluation.

The need for systematic evaluation of LLMs in Islamic legal contexts is increasingly urgent. While these models have generated interest for their potential to assist with complex legal and religious reasoning (Aldahoul and Zaki 2025), their reliability in such sensitive domains remains largely unexplored (Atif et al. 2025). A particularly critical concern is *hallucination*—a pervasive but not uniformly defined phenomenon where LLMs generate plausible yet factually incorrect, fabricated, or ungrounded information. The very definition of hallucination remains contested and context-dependent (Venkit et al. 2024; Magesh et al. 2025): What constitutes hallucination in general question-answering differs from legal contexts, which in turn differs from religious and jurisprudential domains (Ho 2024; Dahl et al. 2024). In secular legal systems, hallucination might involve fabricated case citations or misrepresented precedents. However, in Islamic legal contexts, hallucination takes on qualitatively

different and more severe forms: fabricated religious rulings (*fatāwā*), misattributed scholarly opinions, incorrectly cited textual evidence, or invented legal principles can distort religious practice, undermine scholarly authority, and produce spiritual and societal consequences distinct from civil legal errors (Imtiaz and Shafique 2019). When Muslims consult LLMs for guidance on matters affecting their spiritual well-being and religious guidance, understanding and detecting hallucination becomes a technical challenge with significant ethical implications.

Current evaluation frameworks face several critical gaps when applied to Islamic legal reasoning. Existing benchmarking efforts are either specialized in single domains (Aldahoul and Zaki 2025) or limited to subsets of jurisprudential schools (Atif et al. 2025). Existing evaluation frameworks are often limited to a single jurisprudential school or a narrow set of legal topics. No evaluation framework spans multiple domains across both Sunni and Shī'a traditions across diverse legal topics. Multi-domain and cross-school evaluation presents distinct challenges: different domains operate under varying epistemic principles (e.g., ritual worship prioritizes textual fidelity while transactions emphasize societal welfare), and different jurisprudential schools apply divergent methodological frameworks to identical source texts, producing legitimately different valid rulings. This requires assessing reasoning coherence within each school's methodology rather than against a single correct answer. Moreover, current approaches inadequately test the sophisticated reasoning capabilities required for authentic Islamic legal analysis, such as identifying underlying legal rationales (*'illah*), applying analogical reasoning to novel scenarios, and navigating the multi-school nature of Islamic law where the same question may have different valid answers depending on methodological approach.

This paper introduces *IslamicLegalBench*, the first pluralist benchmark for evaluating LLM performance across major areas of Islamic jurisprudence. We develop a multi-tiered evaluation framework spanning 1,200 years of Islamic legal tradition, drawing from 37 foundational legal texts across several schools of jurisprudence. Our benchmark comprises 718 manually curated evaluation instances organized into 13 task types across three complexity tiers: Low (bibliographical and factual recall), Moderate (condition enumeration, comparative distinctions, statutory synthesis), and High (legal rationale identification, analogical application, cross-school synthesis, maxim mapping). We conduct a systematic evaluation of 9 state-of-the-art LLMs using both zero-shot and few-shot prompting, employing both human domain-experts and an LLM-as-a-Judge framework for fine-grained assessment across correctness, hallucination, abstention, and false premise detection. Our analysis reveals critical insights into the limitations of current LLMs for Islamic legal reasoning, including the failure of in-context learning, problematic confidence-hallucination patterns, and significant performance disparities between closed and open-source models.

1.1 Contributions

Our work makes the following scholarly contributions that extend the state of the art:

1. **A Novel Multi-School Benchmark for Islamic Law:** We develop a novel evaluation framework that is the first to systematically incorporate seven schools of

Islamic jurisprudence, including both Sunni and Shi'i traditions. Our benchmark is designed to test a wide array of tasks, from foundational recall to complex analogical reasoning.

2. **Discovery of In-Context Learning Failure:** We tested 9 state-of-the-art LLMs and revealed that few-shot prompting provides minimal and inconsistent benefit for Islamic legal reasoning (per-model changes range from -3.64 to +4.16 points, and the strongest systems change by at most +2.23). Our analysis uncovers the mechanism: moderate-complexity tasks requiring accurate knowledge exhibit consistent failure, while high-complexity tasks allow semantic generalization, projecting competence despite lacking precise textual knowledge.
3. **Identification of Critical Safety Vulnerabilities:** We expose risky failure modes including confident incorrectness (correctness <50%, hallucination >25%, abstention <1%), widespread sycophantic behavior (five models accepting false premises >40% under few-shot prompting), and few-shot prompting worsening sycophancy by 3.49 percentage points. We also quantify significant performance disparities between closed and open-source models: 21.73 percentage points in correctness and 20.15 percentage points in hallucination rates.
4. **Benchmark Release:** We release 20% of IslamicLegalBench publicly as a development set, maintaining 80% held-out to preserve evaluation integrity and prevent data contamination.

Together, these contributions provide the first systematic understanding of LLM capabilities and limitations in Islamic jurisprudence, revealing that current models fundamentally lack the Islamic legal knowledge required for reliable deployment and that addressing this gap requires training on comprehensive Islamic legal corpora. Our dataset achieves comprehensiveness through several dimensions: it spans multiple jurisprudential schools, covers a diverse range of legal topics across core domains of Islamic law, and draws from over 1,200 years of Islamic legal scholarship, ensuring temporal breadth. This multi-dimensional coverage captures the methodological diversity and evolutionary depth essential to Islamic jurisprudence (detailed further in Section 3).

1.2 Organization of the paper

The remainder of this paper is organized as follows: Section 2 reviews related work in legal AI benchmarking, Islamic NLP datasets, and scholarly perspectives on AI in Islamic jurisprudence. Section 3 describes our dataset creation process, including the Source Dataset of 271 question-answer pairs from 37 authoritative texts and the Benchmarking Dataset consisting of 718 instances with 13 task types across three complexity levels. Section 4 presents the experimental setup, including the 9 evaluated LLMs and prompting strategies. Section 5 details our evaluation methodology using the LLM-as-a-Judge framework with comprehensive scoring criteria. Section 6 presents results across correctness performance by complexity tier, hallucination patterns, false premise acceptance rates, and in-context learning effectiveness. Section 7 discusses architectural limitations in legal reasoning and their alignment with broader LLM performance patterns. Section 8 addresses limitations and future directions.

2 Related work

This section surveys prior work on LLM evaluation in legal domains, examining both general legal benchmarking and Islamic-specific efforts. We organize our review into three areas: general legal AI benchmarking that provides methodological foundations, foundational datasets and benchmarks in Islamic contexts, and ethical and scholarly perspectives on AI in Islamic jurisprudence. From these strands, we identify critical gaps that motivate our comprehensive benchmark.

2.1 LLM benchmarking in general legal contexts

The intersection of technology and legal reasoning has progressed from early conceptual explorations to systematic empirical evaluation. Susskind's *The Future of Law* (Susskind 1998) introduced the legal academy to the disruptive potential of information technology (IT) in the mid-1990s, arguing that emerging digital systems would profoundly reshape legal practice and the processes of legal reasoning. Nearly three decades later, his work *Tomorrow's Lawyers* (Susskind 2023) revisits these predictions in the context of contemporary artificial intelligence, analyzing how modern AI tools now perform components of legal work once assumed to require exclusively human judgment. Similarly, Grant and Wischik (Grant and Wischik 2020) provide key conceptual foundations for understanding the application of machine learning to law, highlighting how the distinctive features of legal reasoning, i.e., its reliance on precedent, interpretation, and context-sensitive judgment, create methodological challenges for AI systems. Together, these works trace a scholarly trajectory from early technological transformation arguments to contemporary concerns about how AI operationalizes legal tasks.

Building on these conceptual foundations, recent empirical research has systematically evaluated LLMs on legal reasoning tasks, revealing specific limitations. The systematic evaluation of hallucination in legal AI systems informed our benchmark design. The work of hallucination detection in Dahl et al. (2024), followed by Magesh et al. (2025), assesses both general LLMs and specialized legal tools on case law queries, respectively, defining "factual hallucination", a response unfaithful to real-world facts, as a particularly significant error in legal contexts where correct citation and accurate representation of law are essential. This definition shaped our own hallucination detection framework.

The first of these studies (Dahl et al. 2024) demonstrated the value of evaluating models exclusively on their internal knowledge, without retrieval augmentation, an approach we adopt in our evaluation. Its benchmark involved approximately 200,000 queries for four models (GPT-4 (OpenAI 2023), GPT-3.5 (OpenAI 2022), Llama 2 (Meta AI 2023), and PaLM 2 (Google 2023)), using 14 task templates across three complexity levels (Low, Medium, and High), which directly inspired our three-tier complexity structure. This work employed a GPT-4-based "**LLM-as-Judge**" framework to assess outputs, a methodology we adopt using o3, and revealed high mean hallucination rates across all models, underscoring the limitations of LLMs in legal reasoning without retrieval support and motivating our systematic hallucination measurement.

The subsequent study (Magesh et al. 2025) examined dedicated legal tools integrating retrieval-augmented generation (RAG), including Lexis+ AI (Lexisnexis 2023), Westlaw AI-Assisted Research (Reuters 2024b), and Ask Practical Law AI (Reuters 2024a), with GPT-4 (OpenAI 2023) scores as a baseline. It distinguished three outcome categories (Accurate, Hallucinated, Incomplete) and tested four capability categories: general legal research, jurisdiction- and time-specific queries, factual recall, and, notably, **false-premise queries** to probe system robustness. This false-premise testing approach directly motivated our inclusion of false premise detection tasks (see Section 11.2). Their results showed substantial variation: Lexis+ AI achieved 65% accuracy with 17% hallucination, Westlaw AI 42% accuracy with 33% hallucination, and the GPT-4 baseline 49% accuracy with 43% hallucination, demonstrating that even RAG-augmented systems exhibit concerning hallucination rates. Another study finds that, at present, these LLMs struggle with performing legal reasoning tasks but may still be useful in supporting lawyers with drafting legal documents and other related services (Trozze et al. 2025).

The LegalBench benchmark (Guha et al. 2023) provided methodological insights that directly shaped our task design. Drawing from the IRAC (Issue, Rule, Application, Conclusion) methodology (Irac Method 2025), it decomposes legal reasoning into six distinct cognitive types. Evaluation of 20 LLMs revealed three key findings that inform our approach: (1) Task-specific performance variation, validating our need for fine-grained evaluation across 13 task types; (2) Prompt design matters, with description-based prompts often outperforming simple queries, informing our prompt engineering; and (3) High sensitivity to in-context examples, which informed our systematic comparison of zero-shot versus few-shot prompting approaches.

The broader legal-AI literature shows that LLMs struggle in two recurrent ways: they hallucinate factual details and they break down on tasks requiring multi-step legal reasoning or nuanced contextual grounding (Ho 2024). One study found hallucination rates of at least 75% on queries about a court's core ruling (Dahl et al. 2024). A dominant error pattern is **"Wrong Conclusion from False Premises,"** where models accept factually incorrect premises and build plausible but erroneous arguments. This contrafactual bias poses critical dangers in religious contexts where a single mistake can have profound spiritual implications, directly motivating our benchmark's inclusion of false premise detection tasks (see Section 11.2) to test if LLMs can reject incorrect assumptions in Islamic legal queries.

2.2 Foundational datasets and benchmarks in Islamic context

While general legal AI research provided methodological foundations, understanding the landscape of existing Islamic NLP resources revealed critical gaps our benchmark addresses. Research in Islamic NLP has progressed from general-purpose text corpora to specialized legal benchmarks. We trace this evolution through three generations of work: early sacred text corpora, domain-specific legal datasets, and recent task-specific benchmarks for Islamic legal reasoning.

Efforts on Islamic text corpora Early efforts compiled large-scale corpora such as Quran-NLP (QURAN-NLP 2025), encompassing the Qur'an, Hadith, translations,

and *tafsīr* (exegesis) with over 700,000 Hadiths, and the 14 Books Hadiths Collection (Slepovichev 2025), providing over 50,000 bilingual narrations with metadata. While these datasets enable basic NLP tasks such as translation, question-answering, sentiment analysis, and summarization, they lack the jurisprudential and school-specific annotation necessary for rigorous legal analysis (Khair and Sawalha 2025), revealing the need for specialized legal benchmarks. To address these limitations, specialized legal corpora emerged. AraLegal-BERT (Alqaseemi and Souissi 2022), pre-trained on 4.5 GB (13.7 million sentences) of Arabic legal texts, demonstrated improved accuracy on legal tasks. Fatwa datasets from IslamQA (Munajjid 1996) and Islamweb (Islamweb 1998) provide question-answer pairs for training chatbot systems (Namoun et al. 2024). However, single-source fatwa datasets carry inherent bias risks, potentially creating a “legal monoculture” (Ho 2024) that erases Islamic jurisprudence’s diversity, constituting representational harm (Suresh and Guttag 2021), a concern that motivated our multi-school, multi-source approach spanning seven schools of jurisprudence and 37 foundational texts.

Recent task-specific benchmarks With recent AI advances, specialized benchmarks have emerged for Islamic legal reasoning, providing immediate precedents for our work while also revealing limitations we sought to address. The *FiqhQA* benchmark (Atif et al. 2025) evaluates LLMs across four major Sunni schools (Hanafi, Maliki, Shafi’i, Hanbali) using 960 question-answer pairs from the Kuwaiti Fiqh Encyclopedia (Department of Islamic Research and Encyclopedias 2026) in both English and Arabic. FiqhQA introduced *abstention behavior* evaluation, the ability to recognize when not to answer, addressing concerns that LLMs hallucinate even with RAG, a metric we incorporate into our evaluation framework. Using LLM-as-a-judge evaluation (Bai et al. 2024; Liu et al. 2023), GPT-4o achieved the highest accuracy (46% English, 28% Arabic) but showed substantial school-specific variation (Hanafi 56% vs Maliki 37%), likely reflecting training data bias toward sources like <https://www.IslamQA.org> (Munajjid 1996). GPT-4o was less likely to abstain when uncertain, yielding higher rates of confident incorrect answers, while Gemini showed 90% abstention with 1% errors. This underscores that knowing when not to answer can be as important as answering correctly in religious contexts, a finding that informed our inclusion of abstention measurement and our identification of the “danger zone” of confident incorrectness. However, FiqhQA is limited to four Sunni schools and a single domain (*Fiqh al-‘Ibādāt*), excluding Shi’i schools and complex legal reasoning, gaps our benchmark addresses. The ArabicNLP QIAS 2025 Shared Task (Al-Smadi 2025) focuses on Islamic inheritance law (*‘ilm al-mawārīth*) with multiple-choice questions across three difficulty levels. The QIAS 2025 dataset (QIAS 2025), derived from 32,000 fatwas and verified by inheritance law experts, demonstrated that LLMs can achieve high accuracy on complex, rule-based legal reasoning when the domain is highly structured. However, its focus on a single subdomain limits generalizability, motivating our multi-domain approach. MizanQA (Bahaj and Ghogho 2025), a Moroccan-specific dataset with approximately 1,700 multiple-choice questions, covers Maliki jurisprudence alongside French-influenced conventions but focuses on a single school, emphasizes factual recall over complex jurisprudential reasoning, and

lacks cross-school comparative evaluation, limitations our benchmark systematically addresses.

From this extensive literature review, we identify that existing benchmarks exhibit noticeable limitations that our work addresses: most focus exclusively on Sunni schools, target single domains rather than comprehensive legal reasoning, and emphasize factual recall over sophisticated reasoning like identifying *‘illah*, applying *qiyās*, or cross-school synthesis. Our benchmark addresses these gaps by spanning seven schools, multiple legal domains, and three complexity tiers, testing factual knowledge through advanced jurisprudential reasoning. Table 1 compares our IslamicLegalBench with existing Islamic legal benchmarks.

2.3 Ethical and scholarly perspectives on Islamic AI applications

The technical challenges identified in LLM legal reasoning take on heightened significance in Islamic contexts, where errors carry profound spiritual consequences, a reality that shaped our evaluation priorities and interpretation of findings. Con-

Table 1 Comparison of Islamic Legal Benchmarks

Feature	IslamicLegalBench (Ours)	FiqhQA (Atif et al. 2025)	QIAS 2025 (QIAS 2025)	MizanQA (Bahaj and Ghogho 2025)
Dataset Size	718 evaluation instances	960 Q&A pairs	~22,000 MCQs (20k train, 1k val, 1k test)	~1,700 MCQs
School Coverage	7 schools (4 Sunni + 3 Shī‘ī)	4 Sunni schools	Sunni inheritance law focus	Primarily Maliki (with customary/civil-law elements)
Legal Domains	Multi-domain: contract law, family law, property law, etc	Worship only (‘Ibādāt)	Inheritance law	Moroccan legal system (Islamic + civil + customary)
Task Types	QA on 13 tasks incl. recall, synthesis, ‘illah, qiyās, cross-school	QA with abstention on Worship	Multiple-choice questions	Multiple-choice questions (incl. multi-answer)
Complexity	3 tiers: Low / Moderate / High	-	3 difficulty levels	-
Source Material	37 classical texts (1,200 years)	Kuwaiti Fiqh Encyclopedia	IslamWeb fatwas	Moroccan legal exam sources

temporary and classical scholarly consensus affirms that non-human systems cannot replicate the essential qualities required for religious guidance, as qualified human scholars possess attributes that AI fundamentally lacks.

Contemporary and classical scholarly consensus While some scholars cautiously endorse LLMs as supplementary research tools under strict oversight, many maintain strong reservations that informed our assessment of when LLMs are “ready” for deployment. Islamic Religious Authorities across multiple jurisdictions emphasize that independent reasoning (*ijtihad*) and religious rulings must remain exclusively with qualified human scholars. Egypt’s Dar al-Iftā warns that AI applications cannot distinguish between true and false content (Latifi 2024), a concern with which our hallucination findings are directionally consistent. Similarly, Indonesia’s Nahdlatul Ulama Council prohibits chatbots from issuing fatwas, noting that AI lacks the wisdom and intuition of human *muftis* (Rahman et al. 2023). Iranian clerical leaders also warn that “robots can’t replace senior clerics” (Futurism 2023).

This scholarly position reflects classical Islamic jurisprudential principles that provide interpretive context for our technical findings. Ibn Qayyim al-Jawziyyah argued that failure to understand context is an error, as the same legal principle can yield different outcomes depending on circumstances (Ibn Qayyim al-Jawziyyah 1991), a test AI systems often fail, as our false premise detection results demonstrate. Al-Nawawī highlighted that qualified jurists must be aware of legitimate differences of opinion to provide balanced rulings (Al-Nawaw 1996), AI models risk oversimplifying issues or promoting single viewpoints. Ibn ‘Abd al-Barr articulated that a mufti must embody moral integrity (*‘adālah*) and commitment to justice (‘Abd al-Barr 1994), reflecting theological concerns about LLMs’ inability to possess God-consciousness (*taqwā*), sincere intention (*niyyah*), or authentic understanding of individual circumstances, all essential for Islamic guidance.

Scholars permitting limited LLM use emphasize specific safeguards that inform our recommendations. Islamic AI systems must base themselves on authentic sources (Qur’an, Prophetic tradition (*Sunnah*), scholarly consensus (*ijmā’*), analogical reasoning (*qiyās*)) with strict human oversight (Zakaria et al. 2023), requirements our benchmark helps assess. Saudi Arabia’s Fatwa Robot operates exclusively on pre-verified fatwas, automatically connecting users to live scholars when questions exceed its parameters (Saudi Press Agency 2025), an approach our abstention metrics support. Malaysian scholars recommend AI systems specify their jurisprudential tradition or present multi-sourced answers to avoid doctrinal bias (Mohammed 2024), guidance that motivated our multi-school evaluation framework. The consensus is clear: AI should supplement scholars by providing research assistance and information retrieval, but must never substitute their judgment, wisdom, or authority. The International Islamic Fiqh Academy emphasizes that AI-generated outputs on religious matters must undergo vetting by qualified Islamic scholars (*‘ulamā’*) (International Islamic Fiqh Academy 2025), a position our findings on hallucination rates and false premise acceptance strongly support.

3 Dataset creation

Our dataset creation involved two stages. First, we built a foundational Source Dataset by manually extracting question-answer pairs directly from authoritative texts of Islamic jurisprudence. Second, we developed the Benchmarking Dataset by systematically transforming these source entries into a structured evaluation framework. This framework, organized across three complexity levels, enables a rigorous assessment of an LLM's Islamic legal reasoning capabilities

3.1 Source dataset

The Source Dataset is the result of an expert-led curation process. We began by surveying 60 foundational books of Islamic jurisprudence, focusing on core legal topics including family law, contracts, torts, criminal law, and finance. From this initial pool, we narrowed our selection to the 37 most relevant texts to create a diverse and historically significant collection.

Our reduction from 60 surveyed works to 37 selected texts followed explicit inclusion and exclusion criteria designed to balance doctrinal authority, domain coverage, temporal breadth, and evaluation integrity. We included works that meet four criteria: (1) **authoritative standing within a school**, reflected in their canonical use in *uṣūl* and *fiqh* instruction and their frequent citation in later juristic writing; (2) **coverage of core substantive domains** targeted by this benchmark (contracts and commercial transactions, family law, property, torts, criminal law, and finance), ensuring that no single domain drives the source pool; (3) **temporal representativeness** across the classical, post-classical, and modern periods to capture doctrinal continuity and change over roughly 1,200 years; and (4) **genre diversity**, combining concise manuals (*mutūn*), extended commentaries (*shurūḥ*), juristic compendia, *fatwā* collections, and codificatory texts, so that questions probe both rule statements and explanatory reasoning. We excluded works that are (a) narrowly limited to ritual practice only, (b) largely duplicative of another selected text within the same school and domain, or (c) so widely circulated in modern web corpora that they posed a higher contamination risk, unless their doctrinal role was indispensable for cross-school comparability. This selection approach aimed to preserve breadth while keeping the benchmark sensitive to school-specific doctrine and method.

Qualitative bias audit: The resulting distribution across schools partly reflects structural constraints in digitization and publication availability. Larger Sunni schools have broader digitized textual traditions, while some Shi'i and Zāhiri works are less available in machine-readable form, which limits the feasible pool. Within these constraints, we adopted a minimum-coverage principle for each school, ensuring representation of all seven traditions and multiple legal domains per tradition, rather than allowing selection to track availability alone. We therefore treat the current school distribution as a pragmatic approximation of the accessible historical record, not as a claim about the relative doctrinal importance of any school. Future expansions will prioritize additional Shi'i and minority-school sources as more texts become digitally accessible.

The seven traditions represented in this benchmark do not exhaust Islamic legal scholarship. Ismaili jurisprudence, Ibādī jurisprudence, and other schools and sub-

traditions each carry rich and continuing interpretive corpora, and their omission reflects no judgment on their doctrinal standing or scholarly significance. The constraint stems from our sampling frame. Our inclusion criteria require, for each school, several distinct authoritative works spanning multiple substantive domains and historical periods, so that questions can probe within-school doctrinal development and support genuine cross-school comparison rather than a single-text entry per tradition. This frame fits schools with broad, multi-work positive-law literatures, as reflected in our per-school text counts (Table 2). It fits less naturally where a tradition's substantive law is concentrated in one foundational compendium: Ismaili positive law, for example, is centered to an unusual degree on al-Qāḍī al-Nu'mān's *Da'ā'im al-Islām*, the officially promulgated Fatimid code that remains the principal legal reference for Tayyibī Ismaili communities. Rather than represent such traditions through a single text, which would not exercise the comparative and enumerative tasks that define this benchmark, we prioritized mainstream Sunnī and Shī'ī schools with multi-work coverage while deliberately retaining minority representation (Zāhirī, Zaydī). Extending the benchmark to Ismaili, Ibādī, and further traditions, with sampling strategies suited to their textual structure, is a stated priority for future work (Section 9).

This final set of 37 texts, spanning over 1,200 years of legal tradition (8th century to contemporary period), yielded 271 expert-extracted question-answer pairs. The collection reflects both historical significance and methodological diversity across seven schools of jurisprudence: Hanafi (10 works, 26%), Maliki (8 works, 21%), Hanbali (6 works, 16%), Shafi'i (5 works, 13%), Twelver Shi'i (4 works, 11%),

Table 2 Overview of Foundational Sources by School of Law

School	# Books	Key Examples (Title - Author)
Hanafi	9	Mukhtasar al-Quduri - <i>Quduri</i> ; al-Mabsut - <i>Sarakhsi</i> ; al-Fatawa al-Hindiyyah (al-'Alamgiriyyah) - <i>Nizam</i>
Maliki	8	Al-Muwatta' - <i>Malik</i> ; Mukhtasar Khalil - <i>Khalil</i> ; Al-Qawānīn al-Fiqhiyyah - <i>Ibn Juzayy</i>
Shafi'i	5	Al-Umm - <i>Shafi'i</i> ; Al-dHāwī al-Kabīr - <i>Mawardi</i> ; Fatāwā - <i>Ghazali</i>
Hanbali	6	Al-Mughnī - <i>Ibn Qudāmah</i> ; Al-Inṣāf fī Ma'rifat al-Rājih - <i>Mardawi</i> ; Mukhtasar al-Khiraqī - <i>Khiraqī</i>
Zahiri	2	Al-Muḥallā bi-al-Āthār - <i>Ibn Hazm</i> ; Al-Iḥkām fī Uṣūl al-Aḥkām - <i>Ibn Hazm</i>
Zaydi	3	Al-Azhār fī Fiqh al-A'imma al-Aṭhār - <i>al-Murtaḍā</i> ; Sharḥ al-Tajrīd - <i>al-Murtaḍā</i> ; Sharḥ al-Tajrīd - <i>Harūnī</i>
Twelver (Ja'fari)	4	Al-Kāfi - <i>Kulaynī</i> ; Tadhkirat al-Fuqahā' - <i>al-dHillī</i> ; Taḥrīr al-Wasīlah - <i>Khomeini</i>

Zaydi (3 works, 8%), and Zahiri (2 works, 5%) . This distribution is intended to reflect the relative availability of digitized works and the extensive textual traditions of the larger Sunni schools; by comparison, legal sources for schools like the Zahiri are significantly rarer. Table 2 contains the number of Islamic authoritative books belonging to the 7 schools of thought used in the construction of our source dataset.

The complete list of all 37 texts with full author names, titles, and historical periods is provided in Appendix A.

The works represent diverse legal genres, including systematic manuals (*mutūn*), detailed commentaries (*shurūh*), collections of legal edicts (*fatāwā*), and legal codes. Each of the 271 entries includes comprehensive metadata: source text identification (author and book title), jurisprudential school, historical period, textual references, and complexity assessments.

3.1.1 Data collection process

A key consideration during selection was mitigating data contamination. This problem occurs when a model is evaluated on texts it has already “memorized” from its training data, leading to artificially inflated performance scores that do not reflect genuine reasoning. To minimize this risk, we made a deliberate effort to select authentic legal texts that are less commonly discussed in modern web corpora or are not widely available in easily machine-readable formats. While no selection process can perfectly guarantee non-exposure, this approach helps ensure our benchmark is genuinely testing the model’s ability to reason about Islamic legal principles rather than its capacity to recall memorized text.

A related but distinct contamination risk arises after release: publicly posted benchmark items are crawled and can enter the training corpora of future model versions, letting later systems answer by recall of the test items rather than by knowledge or reasoning, and silently invalidating longitudinal comparison (Sainz et al. 2023; Zhao et al. 2025). Documented contamination in widely distributed benchmarks reaches 45% (Xu et al. 2024). We therefore release 20% of IslamicLegalBench as a public development set while maintaining the remaining 80% as a held-out test set evaluated only through leaderboard-based submission, in which participants submit model predictions and never receive the test items directly, consistent with closed-test practice in contemporary benchmarking (Zhao et al. 2025; Rajore et al. 2024).

3.1.2 Dataset creation workflow

Figure 1 illustrates our systematic two-stage process. In the **initial data collection stage**, an Islamic law expert (associate professor at College of Law with expertise in Islamic Law and Jurisprudence) carefully¹ read through all 37 foundational books to extract and formulate question-answer pairs, producing the source dataset of 271 curated entries.

¹ The curating expert is male, holds a Ph.D. in Comparative Islamic Law, and has over 20 years of experience in Islamic legal scholarship, with an extensive publication record in Islamic jurisprudence, Islamic ethics, and *maqāṣid al-sharīʿa*. His training is comparative across the *madhāhib* rather than affiliated with a single school, which reduced the risk of anchoring item selection and phrasing to one tradition’s interpretive assumptions.

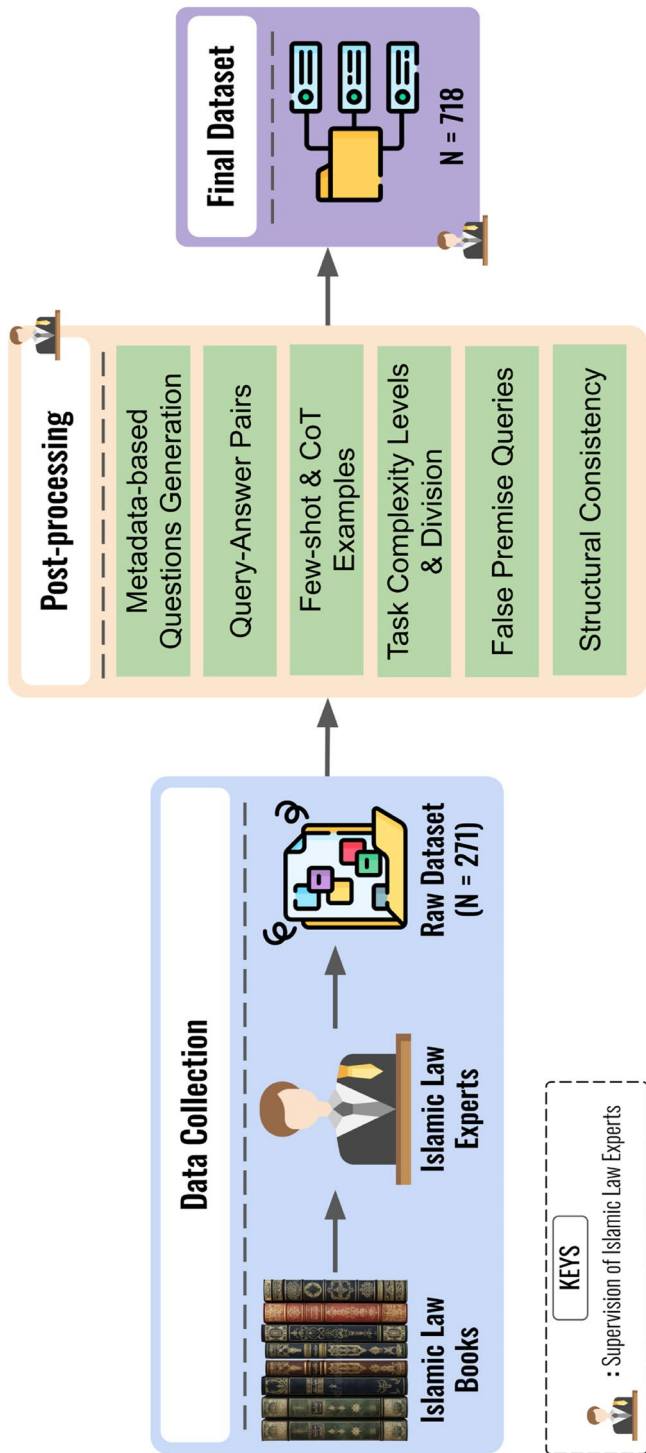


Fig. 1 Comprehensive workflow of dataset creation from source Islamic legal texts to final benchmarking dataset, showing the transformation from 271 expert-curated source entries to 718 structured evaluation items under continued expert supervision

This manual extraction ensures authenticity and accuracy in representing Islamic legal principles across different schools and historical periods. Since this stage involves direct extraction from authoritative sources, inter-annotator agreement was not required.

The **post-processing stage** transforms these 271 source entries into 749 evaluation items under the continued supervision of the same Islamic law expert. Out of these 749 items, 31 were held out for few-shot prompting (selecting representative examples across all task types and complexity levels), resulting in a final dataset of 718 evaluation items. This transformation involves: (1) generating metadata-based questions to test bibliographical knowledge; (2) deriving multiple task variants from single source entries; (3) creating few-shot examples for model reasoning guidance; (4) organizing tasks into three complexity levels based on expert judgment of required jurisprudential sophistication; (5) introducing false premise queries to test whether LLMs identify and correct wrong assumptions; and (6) ensuring structural consistency across all evaluation items.

Complexity Level Assignment: The classification of tasks into Low, Moderate, and High complexity tiers was determined through expert assessment based on the depth of jurisprudential knowledge and reasoning required. Low-complexity tasks involve direct recall of bibliographical facts or basic legal principles. Moderate-complexity tasks require synthesis across multiple conditions, comparative analysis between rulings, or enumeration of structured legal requirements. High-complexity tasks demand sophisticated reasoning, including analogical application of legal principles, cross-school synthesis, identification of underlying legal rationales, or mapping to abstract legal maxims. While this classification involves expert judgment (making it inherently subjective), it is grounded in established frameworks of Islamic legal pedagogy where students progress from memorization (*hifz*) to comprehension (*fahm*) to application (*taṭbīq*) and finally to synthesis and derivation (*istinbāt*).

3.2 Benchmarking dataset

The benchmarking dataset serves as the primary evaluation framework for systematically assessing LLM performance in Islamic legal reasoning. This comprehensive benchmark comprises 718 structured evaluation items that rigorously test model capabilities across multiple dimensions of jurisprudential understanding. Unlike traditional question-answering datasets, our benchmark is specifically designed to probe both surface-level knowledge retrieval and deep legal reasoning abilities required for authentic Islamic jurisprudential analysis. We organize evaluation tasks into three complexity levels (see Fig. 2 for visual taxonomy), each containing distinct task families that progressively challenge different cognitive capabilities:

- *Low-Complexity:* Questions assess basic factual recall and concept existence within a text, testing the model's core grounding.
- *Moderate-Complexity:* Questions require the precise enumeration of legal conditions or the synthesis of specific rules, testing for accuracy and the ability to avoid extraneous information.
- *High-Complexity:* High-Complexity questions probe the model's capacity to analyze and articulate the underlying jurisprudential reasoning ('illah), compare legal principles, and apply them to novel scenarios.

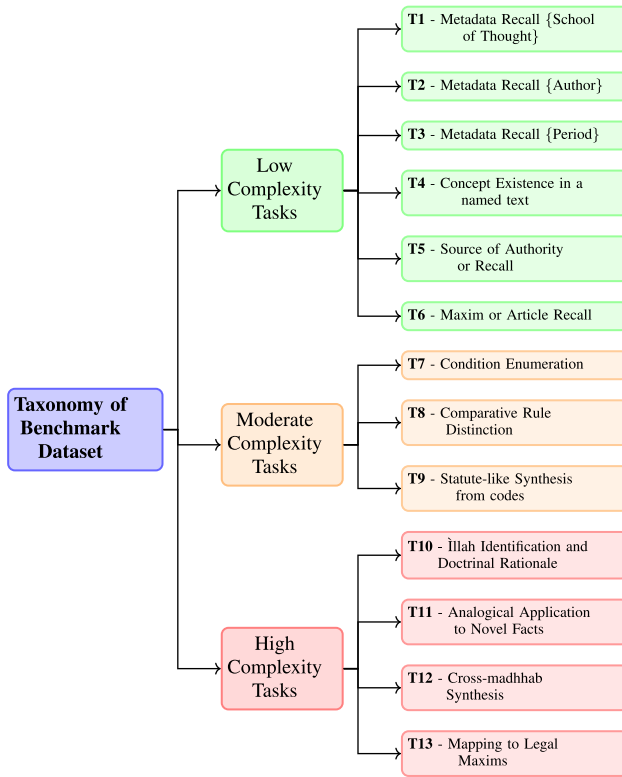


Fig. 2 Taxonomy of benchmark dataset tasks (T1 till T13) organized by complexity levels

Our finalized benchmarking dataset comprises 718 evaluation items distributed across 13 task types spanning three complexity levels. Low-complexity tasks (T1-T6) constitute the majority with 509 instances (71%), moderate-complexity tasks (T7-T9) account for 112 instances (15%), and high-complexity tasks (T10-T13) include 97 instances (14%). This distribution reflects our dataset creation process: while multiple low-complexity tasks (like T1-T4 metadata and concept recall) could be derived from the metadata of a single source entry, the moderate and high-complexity tasks required unique, manual formulation by Islamic law experts for each specific legal scenario. Table 3 presents each task family with representative query examples and the number of instances (N) in our dataset. Notably, concept existence verification (T4) represents the largest single category with 267 items, reflecting the fundamental importance of grounding verification in Islamic legal texts.

In addition to the primary benchmark tasks, we designed a small set of *false premise variants*. These queries deliberately introduce incorrect assumptions (e.g., misattributed authorship, wrong historical period, or fabricated conditions) in the queries with respect to the context. The purpose of these false premise queries (FPQs), thus showing the knowledge and reasoning capability that they possess, instead of confidently accepting the false belief and then fabricating answers. These FPQs are

Table 3 Task families, query examples, and numbering across three complexity levels

Complexity	ID	N	Task family	Query template examples
Low	T1	34	Metadata recall (School)	“What school of thought is associated with Mukhtasar al-Qudūrī?”
Low	T2	33	Metadata recall (Author)	“Who is the author of Mukhtasar al-Qudūrī?”
Low	T3	34	Metadata recall (Period)	“What historical period is Mukhtasar al-Qudūrī from? State the century.”
Low	T4	267	Concept existence in a named text	“Does Mukhtasar al-Qudūrī by Abū al-Ḥasan Aḥmad ibn Muḥammad al-Qudūrī treat Khiyaar al-‘ayb? Answer Yes/No, cite page if Yes.”
Low	T5	83	Source of authority/recall	“Does the author mention any other jurist’s opinion regarding the permissibility of manufacturing contracts (istisnā’)? Answer Yes/No, and if yes, who?”
Low	T6	13	Maxim or article recall	“State Majalla Article 167’s contract-formation rule in one line.”
Moderate	T7	28	Condition enumeration (shurūt)	“List the conditions for a valid salām as held by Abū dHanīfa in the cited page.”
Moderate	T8	21	Comparative rule distinction	“Do conditions differ for gold-for-gold vs gold-for-silver? Give one-sentence reason.”
Moderate	T9	57	Statute-like synthesis from codes	“Synthesize Article 182’s effect when off-topic speech intervenes between ijāb and qabūl.”
High	T10	63	‘Illah identification and doctrinal rationale	“Explain the ‘illah for invalidating ‘one of two garments’ in al-Mabsūt and connect to munāza‘ah.”
High	T11	12	Analogical application to novel facts	“Apply mā laysa fī milkīh to selling a car intended to be bought next week.”
High	T12	5	Cross-madhhab synthesis	“Contrast Abū dHanīfa vs Abū Yūsuf on re-delegation in muḍāraba and justify the mashhūr rule.”
High	T13	17	Mapping to legal maxims	“Given bay‘ bi-al-wafā’, explain why the Majalla treats intent over form and identify the operative maxim.”

designed in a way to test whether pre-trained models use their internal knowledge to rebut these falsely plausible questions and correct the wrongly stated question (Hu et al. 2023). We mark these tasks with a prime symbol (T1’, T2’, T3’, T7’). Detailed explanations of the false premises in these examples are provided in Section 11.2.

3.2.1 Construction and evaluation methods

Benchmark entries were systematically derived from source questions through template-based instantiation and editorial rewriting to reduce ambiguity. Each task underwent quality review for clarity, citation accuracy (where applicable), and consistency of complexity labels to ensure alignment with the intended cognitive demands. Recent evaluations of LLMs in legal-adjacent domains show that models achieve high accuracy on structured formats such as true/false and multiple-choice questions. However, their performance drops considerably on open-ended questions that require applying rules and reasoning through a scenario, indicating continuing limitations in complex legal task execution (Gogani-Khiabani et al. 2025). Hence, our benchmark also focuses on single-turn prompt–response questions that demand

precise legal interpretation and grounded justification, offering a realistic assessment of LLM capabilities on Islamic legal reasoning (Table 4).

We used reference-based evaluation across all task types: model outputs were scored by an LLM-as-Judge using an Exact Semantic Meaning (ESM) rubric that directly compares outputs to ground-truth (gold) reference answers. The ESM criterion isolates the core elements required by the references while permitting variation in writing style, formatting, and phrasing, thereby enabling objective scoring via direct comparison with authoritative references. For high-complexity tasks requiring reasoning, we evaluated only the final responses; intermediate reasoning was not scored because the dataset provides final answers but no reference chains of thought. Any reasoning traces produced by models were supplied to the LLM-as-Judge as context, not as inputs to the scoring criteria.

4 Setup

This section describes our experimental setup, including the evaluated LLMs, prompting strategies (zero-shot and few-shot in-context learning), and abstention handling protocols. We provide comprehensive implementation details to ensure reproducibility and transparency.

Table 4 False premise variant tasks

Complexity	ID	N	Task family	Query template examples
Low	T1'	15	False premise detection (School)	"Analyzing the major Maliki work 'Badā'i' al-Ṣanā'i' fi Tartīb al-Sharā'i'", what distinctive systematization methods does it employ that exemplify Maliki approaches to organizing legal principles?"
Low	T2'	15	False premise detection (Author)	"How did al-ʿAllāmah al-dHillī organize the hadith material in his work 'Al-Kāfi'?"
Low	T3'	15	False premise detection (Period)	"As 'Alā' al-Dīn al-Kāsānī wrote 'Kitāb al-Mabsūṭ' in the 20th-century CE Modern period, how does it address contemporary Islamic legal challenges?"
Moderate	T7'	6	False premise detection (Condition enumeration)	"According to the ruling on page 145, a marriage contract is formed by offer (ījāb) and acceptance (qabūl) expressed in terms that signify marriage, and it is not valid unless it is witnessed by four free, adult, sane, and Muslim men. What happens if only three witnesses are present?"

4.1 LLMs

LLMs are widely accessible and increasingly used for everyday question answering, including by Muslim users seeking religious guidance. Although these public platforms are not specialized for the nuances of Islamic legal reasoning, their ubiquity makes rigorous evaluation essential. Because LLMs are deployed through widely used interfaces (e.g., ChatGPT, Gemini, Grok) and accessed by both lay users and developers via APIs to build domain-specific applications, any shortcomings can propagate at scale. Benchmarking frontier models is therefore a prerequisite for safe, reliable, and accountable deployment. Although some specialized legal models can do better than general models on specific legal tasks (Guo et al. 2025), these systems are still uncommon or niche-specific and not used by the general public. For this reason, for the scope of this study, we focus on mainstream general-purpose LLMs. These are the models that set the overall performance standard, power the most popular platforms, and are the ones people, including Muslims, usually rely on in daily life for guidance on various matters. Evaluating these widely used systems is therefore more important for understanding real-world risks.

To ensure coverage of the state of the art, we include the latest closed-source commercial models and access them directly through their official APIs (e.g., OpenAI, Google, Anthropic, and xAI). Alongside these, we also benchmark open-source frontier models deployed via Together AI (Together AI 2025), which provides scalable access to open-source models. Table 5 provides an overview of the evaluated off-the-shelf LLMs, listing their companies, model names, source types, providers, and model sizes where publicly available.

Table 5 Overview of Evaluated Off-the-Shelf LLMs

Company	Model Name	Model API Version	Type	API Provider	Model Size
Anthropic	Claude Sonnet 4.5 (Anthropic 2025)	claude-sonnet-4-5-20250929	Closed Source	Anthropic	Unknown
Google	Gemini 2.5 Pro (Deepmind 2025)	gemini-2.5-pro	Closed Source	Google	Unknown
OpenAI	GPT-5 (OpenAI 2025a)	gpt-5	Closed Source	OpenAI	Unknown
xAI	Grok-4 (xAI 2025)	grok-4-0709	Closed Source	xAI	Unknown
Meta	Llama 4 Maverick (Meta AI 2025)	meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8	Open Source	Together AI	400B (17B Active)
Meta	Llama 3.1 (Meta AI 2024)	meta-llama/Meta-Llama-3.1-8B-Instruct-Turbo	Open Source	Together AI	8B
OpenAI	GPT-OSS-120B (OpenAI 2025b)	openai/gpt-oss-120b	Open Source	Together AI	120B
DeepSeek	DeepSeek R1 (Deepseek 2025)	deepseek-ai/DeepSeek-R1	Open Source	Together AI	671B
Qwen	Qwen3 (Qwen Team 2025)	Qwen/Qwen3-235B-A22B-Instruct-2507-tput	Open Source	Together AI	235B

4.2 Prompting techniques

4.2.1 In-context sampling

We employed in-context prompting techniques to evaluate the performance of LLMs in our Islamic legal reasoning benchmarking. To systematically assess model capabilities across different prompting paradigms, we evaluate all tasks across all complexity levels using both *zero-shot* and *few-shot prompting*. This approach allows us to measure how effectively LLMs can perform Islamic legal reasoning with and without the influence of contextual examples, providing insights into their inherent understanding versus their ability to learn from in-context demonstrations.

In *zero-shot prompting*, models respond without illustrative examples, relying solely on their pre-trained knowledge and the task instructions. This setting evaluates a model's unassisted capacity to interpret and answer Islamic legal queries across all complexity levels.

In *few-shot prompting*, we provide exemplars that specify the expected output format and reasoning approach for the task type. These contextual examples help models internalize the task structure and produce more aligned responses. However, few-shot prompting also shapes the model's internal decision process; outputs can be highly sensitive to the provided exemplars, introducing example-induced biases. We discuss the benefits and drawbacks of few-shot prompting, especially for Islamic legal reasoning and for tasks that require exact source knowledge with no tolerance for error, in the Results (Section 6).

Few-Shot Prompting Example - T7

Example 1:

<context>

Book: "Jawāhir al-Kalām fī Sharḥ Sharā'i' al-Islām"

Author: Sheikh Muḥammad dḤasan al-Najafī

School of Thought: Twelver Shi'i (Ja'fari)

Period: Modern (19th Century) (d. 1266 H / 1850 CE)

Page Reference: 23/50

</context>

Query: "What is the ruling on selling weapons to the enemies of the faith (a'dā' al-dīn), and under what condition is it permissible, according to page 50?"

Answer: "It is forbidden to sell weapons to the enemies of the faith if it leads to assisting them against Muslims. It is permissible if there is a truce (hudnah) between them and the Muslims."

Example 2: ... —

Actual Query:

<context>

CONTEXT_HERE

</context>

Query: "QUERY_HERE"

Answer:

4.2.2 Abstention behaviour

Similar to false premise questions where LLMs should rebut incorrect assumptions, abstention behavior addresses scenarios where queries fall outside the model's internal or retrieval-augmented knowledge base, or there is a flawed context. In such

cases, LLMs must recognize when to answer and when to refuse, a safety mechanism known as *selective refusal*. Recent studies show that even frontier LLMs struggle with this capability, failing to identify correct reasons for refusal in more than 50% of cases (Muhammed et al. 2025). This challenge takes on particular urgency in Islamic legal contexts, where incorrect guidance can lead to substantial religious and practical consequences. As discussed in Section 2.3, Muslim scholars have extensively cautioned against unvetted AI systems issuing religious rulings, emphasizing that such systems lack the contextual awareness, moral integrity, and human judgment essential for Islamic jurisprudence. These scholarly perspectives underscore a critical implication for LLM evaluation: abstention in cases of uncertainty should be recognized as responsible behavior rather than system failure.

These scholarly insights underscore that abstention must be recognized as responsible behavior rather than penalized as failure. Recent work from OpenAI shows that LLMs hallucinate precisely because standard training and evaluation procedures reward guessing over acknowledging uncertainty (Kalai et al. 2025). They propose that while hallucinations should be heavily penalized during benchmarking, honest abstention behaviors such as “I don’t know” should receive partial credit. The *FiqhQA* benchmark (Atif et al. 2025) demonstrated this principle empirically. GPT-4o achieved higher accuracy but lower abstention rates, yielding more confident incorrect answers, while Gemini exhibited 90% abstention with minimal errors (1% incorrect). This reveals a fundamental trade-off in Islamic legal AI: knowing when *not* to answer can be as critical as answering correctly.

Given these considerations, allowing LLMs to abstain from answering sensitive Islamic legal questions is preferable to providing potentially incorrect guidance (Atif et al. 2025; Kalai et al. 2025). While excessive abstention is not ideal, prompting models to recognize situations where abstention is appropriate can reduce the risk of harmful misinformation and hallucinations. In our experiments, we include a dedicated evaluation of LLM abstention behavior to assess whether models can appropriately recognize the limits of their knowledge in Islamic legal contexts.

5 Evaluation methodology

Our evaluation employs an approach that focuses on capturing all core elements of the reference answer while allowing for natural variations in expression. This methodology “**Exact Semantic Meaning**” recognizes that in fields like Islamic Law, concepts can be articulated in multiple ways without altering their fundamental meaning or context. This means the formatting, writing style, grammar, etc., all of these things are not considered in our results evaluation, and instead, we solely focused on the core elements that constitute the ground truth answers.

5.1 Scoring methods

LLM-as-a-Judge Framework: We employ an LLM-as-a-Judge evaluation framework using o3 (OpenAI 2025c) as the judge model to systematically assess model responses against ground truth answers. For LLM-as-a-Judge, we configured the o3

with the reasoning effort parameter to “*high*” which allocates the maximum compute for the model to think and break down the task into multiple steps to generate the most accurate results. At the time of this study, OpenAI does not provide the option of changing the temperature for reasoning and the latest GPT models.

This approach leverages o3’s advanced reasoning capabilities to perform semantic comparison and scoring of LLM-generated outputs, enabling scalable and consistent evaluation across our comprehensive benchmark. The judge model evaluates each response according to our strict scoring criteria (detailed in Section 5.3), comparing LLM outputs against reference answers to determine correctness and detect hallucinations. In our internal testing on a subset of the benchmark, o3 performed the best out of 5 other LLMs (GPT-5 (OpenAI 2025a), GPT-5 mini (OpenAI 2025d), GPT-4o (OpenAI 2024), Gemini 2.5 (Cloud 2025), and GPT-OSS-120B (OpenAI 2025b)) when judging the response accurately. Tan et al. (2025) introduced JudgeBench, and in their results, they showed that o3 outperformed all the other frontier LLMs in this task. Figure 3 illustrates an example of this framework applied to a false premise query, demonstrating how the judge model correctly identifies fabricated content and assigns appropriate scores.

All LLM response generations were performed using 0.0 temperature settings for deterministic output, ensuring reproducibility between experiments. Models were evaluated on identical prompts with standardized formatting to maintain consistency. For tasks requiring contextual information, we provided the standardized template format described in Section 11, ensuring all models received identical background information.

Inter-Rater Reliability: To annotate responses for correctness and hallucination, we engaged Islamic law experts to hand-score a subset of model outputs using the rubric defined in Section 5.2. Efficient evaluation of AI-generated text remains an open problem, with inherent trade-offs among internal validity, external validity, replicability, and speed (Hashimoto et al. 2019; Liu et al. 2016; Smith et al. 2022). These trade-offs are especially salient when queries approximate real-world tasks. Because authoritative sourcing is central to legal research and writing, adherence to authority must also be assessed manually to ensure that automated self-checking does not become a confounding driver of performance. Building on prior work in legal LLM evaluation (Dahl et al. 2024; Magesh et al. 2025; Atif et al. 2025), we operationalized precise definitions of correctness and hallucination and, on that basis, specified our scoring methodology in Section 5.2. This approach renders benchmarking both costly (API usage and expert annotation) and time-consuming, a necessary trade-off that should inform future discussions about responsibly integrating AI into legal workflows. The definition of key terms, such as hallucination, remains an active area of debate and requires domain-sensitive adaptation. Prior work in legal reasoning (Dahl et al. 2024; Magesh et al. 2025) provides carefully delineated distinctions that we adopt with modest modifications. In our setting, prompts include a context block (e.g., book title, author, school of thought), supplying authoritative source metadata which excludes text-groundedness as a scoring criterion. Section 5.2 details the exact definitions employed in this study.

To verify that our queries and coding schema are sufficiently precise, we measured inter-rater reliability. All task responses were first scored by an LLM-as-a-Judge. We

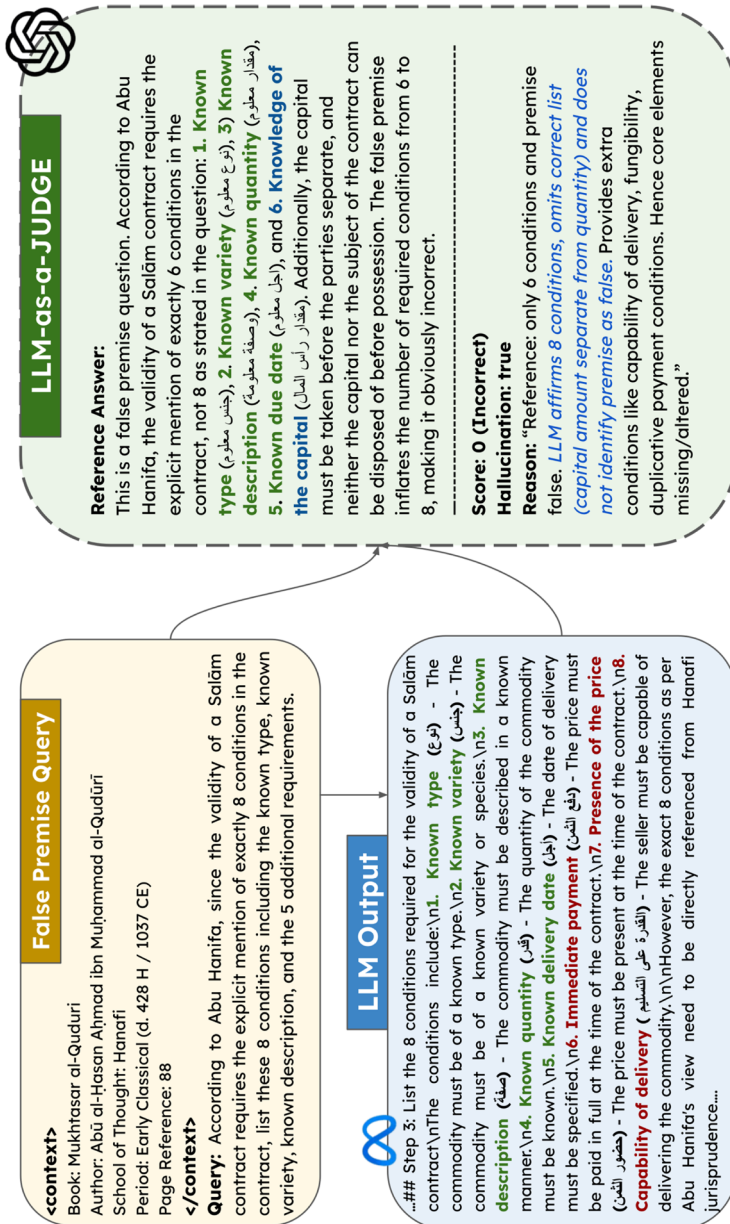


Fig. 3 Illustration of a false premise query evaluated through an LLM-as-a-JUDGE framework. The **LLM Output** (generated by LLaMA-4 Maverick) falsely assumes that Abū dHanifa required eight conditions for a valid *Salām* contract, while the correct jurisprudential reference specifies only six. The output accepts the false premise and hallucinates extra conditions, whereas the judge model correctly identifies the error and assigns a score of 0 (Incorrect). In the 'LLM Output' and 'Reference Answer' panels, **Green** highlights correctly mentioned conditions, **Red** indicates fabricated or hallucinated ones, and **Blue** marks missing but essential elements for required conditions

then drew a stratified sample ($n = 140$) by model, task type, and responses per task from the Moderate and High Complexity tasks (T7–T13), the most technical and reasoning-intensive, and submitted them to two independent Islamic law experts for grading. The experts were blinded to the LLM-as-a-Judge scores and to each other's annotations, and they did not communicate. Their only guidance was our written labeling manual, which we also provided to the LLM-as-a-Judge. Both experts had a good command of the legal reasoning in Islam and knew both English and Arabic languages which was essential in understanding the key terms in the dataset.

Under this protocol, Cohen's kappa (κ) (Cohen 1960) between the two experts was $\kappa = 0.73$ for Scores (refusal: -1 / incorrect: 0 / partially correct: 1 / correct: 2) and $\kappa = 0.72$ for Hallucination, and the mean pairwise Cohen's κ between the LLM-as-a-Judge and the human experts was $\kappa = 0.816$ (see Appendix C). These levels of agreement suggest that the tasks and label taxonomy are well specified, and annotators had a strong agreement. The Section C in the Appendix provides more detailed and specific values regarding reliability testing.

The evaluation prioritizes core element preservation for all essential Islamic legal elements. We award 100% accuracy (score of 2) if all core elements are present and accurately conveyed, even with minor irrelevant mistakes (e.g., a tangential non-core detail that does not impact the core elements). Any core element that is missing, altered in meaning, or misrepresented in semantics/context results in penalization.

5.2 Scoring and hallucination detection

We evaluate responses solely on whether they capture all core elements of the ground-truth/reference answer, disregarding variation in grammar, style, paraphrase, or formatting. Core elements are the essential factual components, principles, and authoritative references from Islamic law that must be present and accurately conveyed without any change in meaning, semantics, or context. The evaluation prioritizes preservation of these elements. A response receives 100% accuracy if all core elements are present and correct, even if it contains minor, non-essential additions (e.g., tangential details that do not affect the core elements). Any omission, semantic alteration, or misrepresentation of a core element results in a penalty as also shown in Fig. 3.

Definition of Hallucination: We define hallucination as any LLM-generated response that is partially correct or incorrect with fabricated or invented content that is either not present or completely diverges from the ground-truth answer. This definition is adapted from (Dahl et al. 2024; Magesh et al. 2025) and is tailored to our setting. In our framework, an Incorrect label typically indicates that the model relied on a fabricated or erroneous assumption at the outset, i.e., a hallucinatory premise, which yields a substantively wrong conclusion. Likewise, a Partially Correct response that includes any invented or misattributed detail is also treated as a hallucination. This includes material that deviates from established Islamic legal sources, adds non-existent information, is fabricated, or misstates jurisprudential principles and authorities. We treat the ground-truth answers as the sole basis for identifying hallucinations, as they specify the decisive elements and underlying rationale. As noted above, our scoring method relies on exact semantic meaning, permitting vari-

ability in phrasing while requiring alignment on core elements. This eliminates the need for multiple variants of the ground-truth answer.

The assessment employs a strict four-tier scoring system with the following methodology:

- **Correct (Score: 2):** Factually accurate, fully relevant to the query and ground truth, with all core elements intact
- **Partially Correct (Score: 1):** Contains some correct information but misses or incompletely addresses key core elements
- **Incorrect (Score: 0):** Contains any factually inaccurate information compared to ground truth or fails to address the query
- **Refusal (Score: -1):** If the model refuses or abstains from answering. This is considered relatively positive behavior since the LLM is refusing to generate information outside of its contextual understanding.

We implement a systematic hallucination-detection procedure to identify responses containing invented, non-existent, or misgrounded elements (e.g., fabricated sources or false attributions other than source text) not present in the ground truth. The hallucination flag is determined as follows: if the response is Incorrect (score 0), the hallucination flag is set to **True**; if the response is Partially Correct (score 1), and it contains invented or misgrounded elements relative to the reference answer set flag to **True** and **False** otherwise; responses that are Correct (score 2) are flagged **False**; Refusal responses (score -1) are not flagged for hallucination, as the model did not generate potentially false or hallucinated content.

5.3 Evaluation process

We employ a structured seven-step evaluation process to ensure consistent, accurate, and reproducible assessment.

1. **Comprehensive Reading:** Evaluators carefully read the query, the ground truth/reference answer, and the LLM response in full to capture the complete context.
2. **Core Element Identification:** All essential factual components are extracted from the reference answer, including Islamic legal principles, authoritative references (e.g., book titles, authors, page numbers), and doctrinal positions across different schools of thought.
3. **Semantic Comparison:** The LLM response is then compared against these core elements to verify their presence and semantic accuracy, with attention to meaning and context rather than grammar, paraphrasing, or stylistic variations.
4. **Non-Core Error Handling:** Minor inaccuracies or tangential details that do not affect the integrity of the core elements are noted but disregarded in the scoring process.
5. **Score Assignment:** Scores are assigned according to a strict rubric: full credit (2) if all core elements are intact and accurately represented, partial credit (1) if only some are correctly captured, no credit (0) if core elements are missing,

- misrepresented, or factually inaccurate, and refusal score (-1) if the model abstains from providing an answer.
6. **Hallucination Detection:** Responses scoring 0 are automatically flagged as hallucinations. For responses scoring 1, evaluators check for fabricated or misgrounded elements and flag hallucination as true if such content is found. Correct responses scoring 2 are never flagged. Refusals with a score of -1 are not flagged either.
 7. **Standardized Output:** Each evaluation is reported in a uniform format for clarity and reproducibility: Score: [0/1/2/-1]. Hallucination: [True/False]. Reason: [concise explanation].

This approach is particularly crucial for Islamic legal reasoning, where the precision of religious and jurisprudential content must be maintained while allowing for the natural linguistic diversity inherent in scholarly discourse.

6 Results

We evaluate nine state-of-the-art LLMs on our Islamic Legal Benchmark, encompassing both closed-source systems and open-source alternatives. Our analysis focuses primarily on few-shot performance, aligning with standard LLM benchmarking practices. Where instructive, we draw upon zero-shot results to illuminate the (in)effectiveness of in-context learning (ICL) specifically for Islamic jurisprudence reasoning. The following subsections present a comprehensive analysis of model performance across multiple dimensions: correctness, hallucination, error patterns, critical reasoning capabilities, and systematic limitations.²

6.1 Correct response performance: The foundation of Islamic legal reasoning

The results reveal substantial heterogeneity in LLMs' capacity for Islamic legal reasoning, with overall correctness ranging from 30.96% to 67.65%. Table 6 and Fig. 4 present the comprehensive breakdown across all 13 tasks and complexity levels.

Three critical findings emerge. First, performance stratifies into distinct tiers: frontier closed-source models (GPT-5, Claude Sonnet 4.5) achieve 65-68% correctness and demonstrate consistent moderate competence across diverse task types, while smaller open-source models (Llama 3.1 8B, GPT-OSS-120B) fall below 37%, exhibiting near-random performance on fundamental Islamic law questions. Mid-tier systems show binary performance patterns, excelling at pattern-matching extraction tasks while failing on genuine jurisprudential reasoning.

² **Note on Performance Tiers and Risk Thresholds.** Throughout this analysis, we categorize models into performance tiers and define risk thresholds (e.g., hallucination >40% as "critical," FPQ acceptance >50% as "dangerous"). These are analytical constructs from empirical observation of our evaluation's performance distribution, established to facilitate meaningful interpretation. They do not represent formally validated cutoffs or industry standards, but serve as heuristic boundaries for characterizing relative model capabilities and safety profiles. Different applications may warrant different thresholds based on deployment requirements, risk tolerance, and domain constraints.

Table 6 Performance comparison of LLMs on our Islamic Legal Benchmark across different task complexities and prompting strategies (Pass@1 rates). Values represent correct response percentages. Best results per task are highlighted in **bold**

Model	Prompt	Low Complexity					Moderate					High Complexity					Overall
		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13			
Claude Sonnet 4.5	Zero-Shot	89.80	87.50	85.71	76.03	36.14	61.54	55.88	33.33	45.61	77.78	83.33	40.00	76.47	65.32		
	Few-Shot	89.80	89.58	87.76	90.64	34.94	53.85	38.24	42.86	56.14	74.60	58.33	60.00	76.47	65.63		
	Zero-Shot	73.47	54.17	59.18	91.39	33.73	38.46	26.47	38.10	38.60	65.08	41.67	40.00	82.35	52.51		
DeepSeek R1	Few-Shot	73.47	58.33	57.14	94.76	26.51	38.46	20.59	47.62	36.84	76.19	58.33	40.00	76.47	54.21		
	Zero-Shot	87.76	85.42	85.71	90.64	38.55	53.85	44.12	42.86	54.39	79.37	50.00	40.00	82.35	64.23		
	Few-Shot	89.80	83.33	87.76	92.51	37.35	46.15	35.29	47.62	52.63	71.43	50.00	40.00	82.35	62.79		
GPT-5	Zero-Shot	89.80	75.00	83.67	85.77	39.76	61.54	44.12	57.14	59.65	82.54	75.00	20.00	76.47	65.42		
	Few-Shot	91.84	83.33	81.63	93.63	46.99	61.54	47.06	57.14	57.89	77.78	50.00	60.00	70.59	67.65		
	Zero-Shot	79.59	77.08	71.43	76.78	38.55	61.54	52.94	47.62	52.63	77.78	33.33	80.00	88.24	64.42		
Grok-4	Few-Shot	81.63	77.08	77.55	91.01	34.94	61.54	32.35	38.10	52.63	74.60	50.00	60.00	70.59	61.69		
	Zero-Shot	71.43	41.67	53.06	98.50	28.92	53.85	20.59	23.81	33.33	58.73	41.67	20.00	70.59	47.40		
	Few-Shot	75.51	54.17	46.94	97.00	22.89	61.54	23.53	23.81	31.58	53.97	58.33	0.00	58.82	46.78		
Llama 3.1 8B	Zero-Shot	30.61	14.58	26.53	79.40	20.48	23.08	2.94	14.29	15.79	28.57	33.33	0.00	58.82	26.80		
	Few-Shot	26.53	20.83	22.45	97.75	19.28	15.38	8.82	14.29	19.30	34.92	50.00	20.00	52.94	30.96		
	Zero-Shot	34.69	12.50	32.65	49.81	31.33	53.85	11.76	23.81	36.84	50.79	41.67	40.00	52.94	36.36		
GPT-OSS-120B	Few-Shot	32.65	16.67	34.69	79.03	22.89	53.85	11.76	19.05	35.09	47.62	25.00	0.00	47.06	32.72		
	Zero-Shot	61.22	43.75	42.86	87.27	31.33	46.15	23.53	33.33	45.61	66.67	41.67	20.00	76.47	47.68		
	Few-Shot	63.27	47.92	32.65	96.63	31.33	46.15	20.59	28.57	35.09	66.67	50.00	40.00	76.47	48.87		

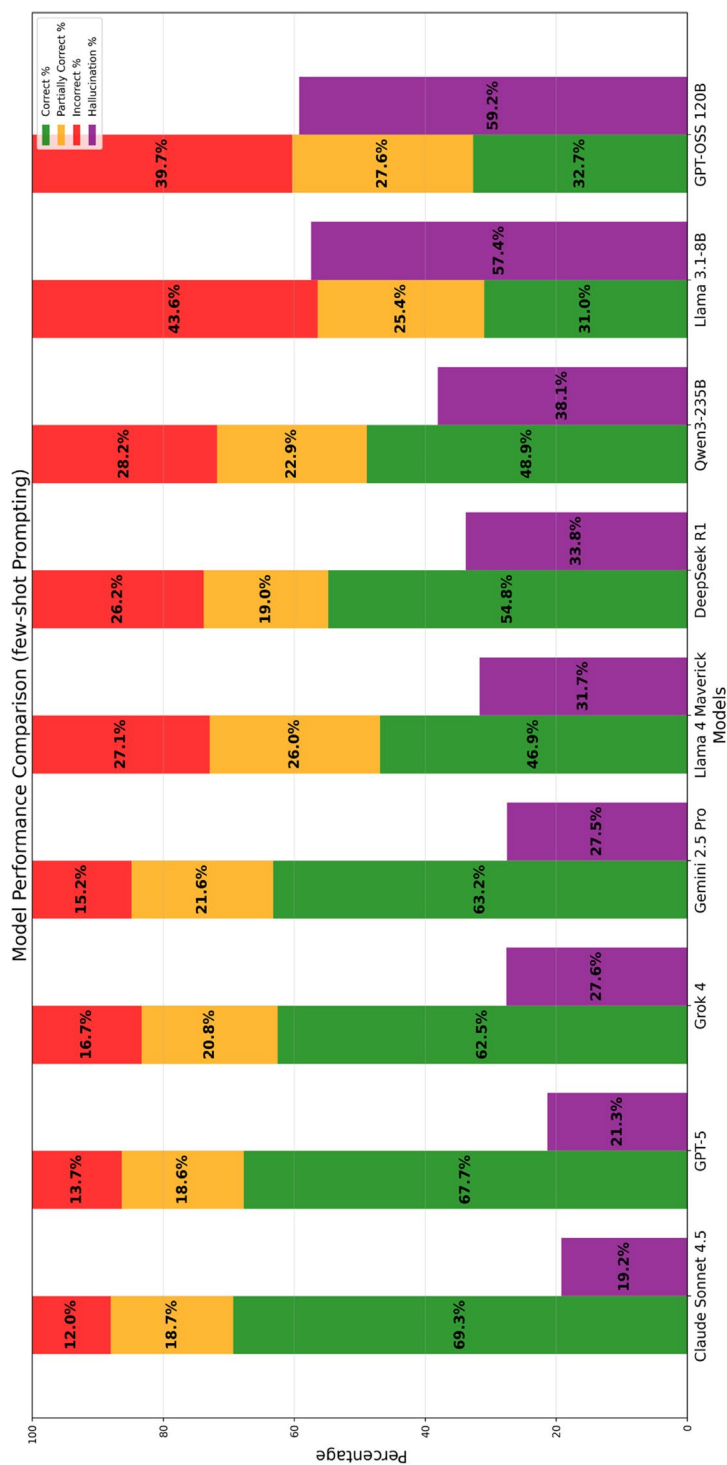


Fig. 4 Overall performance comparison of evaluated LLMs under few-shot prompting. Models are ordered by category (closed-source followed by open-source). The stacked bars decompose responses into correct (green), partially correct (orange), and incorrect (red) categories, while separate purple bars indicate hallucination rates. Performance ranges from 30.96% (Llama 3.1 8B) to 67.65% (GPT-5), revealing substantial variation in Islamic law reasoning capabilities

Second, task difficulty exhibits a paradoxical non-monotonic relationship with human expert-designated complexity. Tasks requiring verbatim knowledge retrieval prove widely challenging: Source Attribution (T5) achieves only 23–47% correctness even among leading models, while Condition Enumeration (T7) and Comparative Distinction (T8) similarly constrain all systems. Conversely, certain high-complexity tasks elicit stronger performance than moderate-complexity ones, as models leverage semantic generalization about legal principles without requiring precise textual knowledge.

Third, this difficulty pattern suggests a fundamental architectural limitation: models demonstrate sophisticated reasoning capabilities when explaining broad legal principles (‘illah identification, analogical application, maxim mapping) but evidently show a drop in performance when Islamic jurisprudence demands condition-specific knowledge from classical fiqh texts. The detailed task-level breakdown in Table 6 shows that extraction tasks (T4: 76–98% correctness) and principle-based reasoning (T10, T13: 70–88%) leverage existing model strengths, while precise enumeration and statute synthesis represent a consistent drop in performance across architectures.

6.2 Hallucination: The shadow side of confident incorrectness

While correctness establishes baseline capability, hallucination rates reveal the critical safety profile of models for real-world Islamic advisory applications. Table 7 and Fig. 5 present detailed statistics and severity patterns across task complexities.

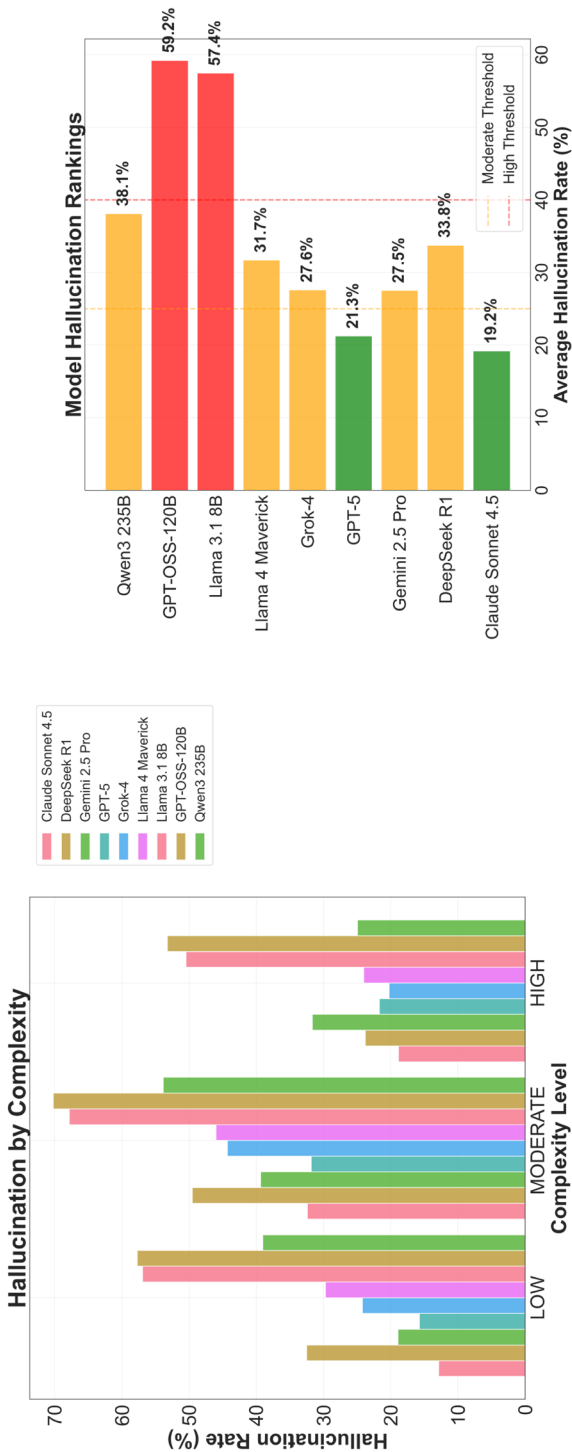
The analysis shows a trimodal safety distribution with implications for deployment. Claude Sonnet 4.5 achieves the strongest safety profile at 19.18% hallucination, with GPT-5 remaining below 25% and Grok-4 near the boundary (23.95% zero-shot, 27.61% few-shot), together forming a relatively secure tier suitable for supervised deployment. A middle tier, comprising Gemini 2.5 Pro, DeepSeek R1, Llama 4 Maverick, and Qwen3 235B exceeds 27–38% hallucination rates, indicating that roughly one-third of responses contain fabricated Islamic legal content and require substantial expert oversight. The bottom tier (Llama 3.1 8B, GPT-OSS-120B) exhibits hallucination above 55%, where more than half of all responses actively contain inaccurate religious content that precludes real-world deployment.

Task-level patterns disclose the underlying mechanism for this behavior. Moderate-complexity tasks requiring precise textual knowledge trigger the highest hallucination: Condition Enumeration (T7) and Comparative Rule Distinction (T8) average 50–70% fabrication rates across models, while Source Attribution (T5) reaches 24–64%. When models lack exact knowledge of legal conditions or juristic citations from classical texts, they tend to fabricate plausible-sounding but incorrect requirements rather than acknowledging uncertainty.

Significantly, this pattern mirrors the non-monotonic difficulty observed in correctness analysis. High-complexity tasks permitting flexible reasoning (T13: Maxim Mapping) elicit minimal hallucination despite their difficulty, while moderate-complexity tasks demanding knowledge accuracy (T7, T8) produce substantially higher fabrication rates. Models can reason extensively about abstract legal principles without hallucinating, but invent precise conditions when lacking training expo-

Table 7 Hallucination rates of LLMs on our Islamic Legal Benchmark across different task complexities and prompting strategies (Pass@1 rates). Values represent hallucination percentages. Lowest (best) results per task are highlighted in **bold**

Model	Prompt	Low Complexity					Moderate					High Complexity					Overall
		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13			
Claude Sonnet 4.5	Zero-Shot	6.12	12.50	12.24	13.86	33.73	15.38	38.24	47.62	35.09	14.29	8.33	60.00	5.88	23.33		
	Few-Shot	8.16	10.42	10.20	8.61	24.10	15.38	47.06	23.81	26.32	12.70	16.67	40.00	5.88	19.18		
	Zero-Shot	24.49	41.67	32.65	8.61	51.81	30.77	64.71	52.38	47.37	19.05	33.33	60.00	0.00	35.91		
DeepSeek R1	Few-Shot	22.45	37.50	38.78	4.87	53.01	38.46	67.65	47.62	33.33	15.87	33.33	40.00	5.88	33.75		
	Zero-Shot	10.20	12.50	12.24	8.99	43.37	30.77	47.06	47.62	28.07	14.29	41.67	60.00	5.88	27.90		
	Few-Shot	8.16	12.50	12.24	7.49	42.17	30.77	47.06	42.86	28.07	19.05	41.67	60.00	5.88	27.53		
GPT-5	Zero-Shot	6.12	22.92	14.29	13.11	43.37	15.38	35.29	42.86	29.82	9.52	8.33	20.00	5.88	20.53		
	Few-Shot	6.12	14.58	18.37	6.37	40.96	7.69	35.29	28.57	31.58	15.87	25.00	40.00	5.88	21.25		
	Zero-Shot	18.37	16.67	22.45	20.60	37.35	7.69	41.18	38.10	29.82	17.46	41.67	20.00	0.00	23.95		
Grok-4	Few-Shot	16.33	22.92	20.41	7.87	46.99	30.77	58.82	47.62	26.32	15.87	25.00	40.00	0.00	27.61		
	Zero-Shot	24.49	50.00	40.82	1.50	56.63	15.38	52.94	33.33	31.58	15.87	16.67	60.00	0.00	30.71		
	Few-Shot	20.41	39.58	46.94	3.00	53.01	15.38	52.94	42.86	42.11	11.11	25.00	60.00	0.00	31.72		
Llama 3.1 8B	Zero-Shot	69.39	79.17	69.39	17.98	66.27	46.15	88.24	71.43	71.93	53.97	58.33	80.00	29.41	61.67		
	Few-Shot	69.39	75.00	77.55	1.87	63.86	53.85	73.53	66.67	63.16	50.79	41.67	80.00	29.41	57.44		
	Zero-Shot	61.22	85.42	65.31	43.45	60.24	38.46	82.35	61.90	50.88	42.86	50.00	40.00	35.29	55.18		
GPT-OSS-120B	Few-Shot	67.35	83.33	65.31	20.22	63.86	46.15	76.47	76.19	57.89	47.62	50.00	80.00	35.29	59.21		
	Zero-Shot	32.65	54.17	48.98	11.24	54.22	38.46	64.71	66.67	36.84	26.98	41.67	80.00	17.65	44.17		
	Few-Shot	32.65	52.08	61.22	3.00	54.22	30.77	64.71	47.62	49.12	20.63	33.33	40.00	5.88	38.10		



(b) Overall hallucination severity rankings with thresholds at <25% (low), 25–40% (moderate), and >40% (critical).

(a) Hallucination rates by model across Low, Moderate, and High complexity tasks under few-shot prompting.

Fig. 5 Hallucination Severity Analysis. Few-shot prompting reveals substantial variation in hallucination robustness: multiple models exceed the 25% moderate-risk threshold, and several reach critical hallucination levels above 40%, particularly on higher-complexity tasks. Panel (a) shows severity by task complexity, while Panel (b) summarizes overall rankings

sure, a fundamental vulnerability for specialized legal domains where exactness is non-negotiable.

6.3 Error modes: Incorrect responses, partial knowledge, and strategic abstention

Beyond binary correctness and hallucination, the distribution of incorrect responses, partial correctness, and abstention patterns reveals models' uncertainty-management behavior and the granularity of their reasoning. Tables 8 and 9 provide comprehensive statistics.

Incorrect response rates broadly stratify into three profiles that appear to align with the correctness hierarchy. Leading models (Claude Sonnet 4.5, GPT-5, Gemini 2.5 Pro) tend to maintain incorrect rates below 15%, with errors appearing concentrated on specific challenging tasks rather than distributed uniformly. This pattern is consistent with, though not conclusive evidence of, broad competence accompanied by targeted reasoning gaps. In contrast, weaker models show a different pattern: Llama 3.1 8B and GPT-OSS-120B produce incorrect responses on 38-43% of queries, rates that exceed their correctness rates when the partial score is excluded. One possible interpretation is that these models are more prone to generating confidently wrong answers, which could pose risks for religious guidance applications where incorrect rulings may affect the validity of worship practices.

Partial correctness patterns point to a notable trend: tasks requiring attribution or doctrinal distinction (T5: Source Attribution, T8: Comparative Distinction) tend to elicit elevated partial scores across leading models, suggesting these systems may recognize that juristic opinions or doctrinal differences exist but do not always succeed in correctly attributing statements to scholars or characterizing distinctions with full accuracy. This "almost correct" pattern points to a particular risk: outputs may appear plausible and reflect apparent Islamic legal knowledge, yet contain subtle errors that could be difficult for non-experts to identify.

The analysis points to a potential misalignment between model confidence and actual accuracy during abstention. Despite achieving only 65-68% correctness, leading models abstain on fewer than 4% of queries, with several models rarely or never refusing any task. Claude Sonnet 4.5 shows marginally elevated abstention (3.66%, concentrated on T5), though this still appears low relative to the observed knowledge boundaries. Models attempt to answer nearly all queries regardless of apparent knowledge sufficiency, which can result in incorrect or hallucinated content rather than acknowledgment of uncertainty. This pattern may compound the safety risks identified earlier: models do not always signal reduced confidence when incorrect, which could limit users' ability to calibrate trust in individual responses.

6.4 False premise queries: Assessing critical reasoning and sycophantic vulnerability

To evaluate critical reasoning capabilities, we embedded 51 deliberately misleading False Premise Queries (FPQs) within Tasks 1, 2, 3, and 7, testing whether models detect and challenge factually incorrect Islamic law assumptions rather than syco-

Table 8 Incorrect response rates of LLMs on our Islamic Legal Benchmark across different task complexities and prompting strategies (Pass@1 rates). Values represent incorrect response percentages. Lowest (best) results per task are highlighted in **bold**

Model	Prompt	Low Complexity					Moderate					High Complexity					Overall
		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13			
Claude Sonnet 4.5	Zero-Shot	6.12	10.42	12.24	16.48	16.87	0.00	14.71	19.05	12.28	9.52	8.33	40.00	11.76	13.68		
	Few-Shot	8.16	10.42	10.20	8.99	16.87	15.38	14.71	4.76	12.28	7.94	8.33	20.00	17.65	11.98		
DeepSeek R1	Zero-Shot	24.49	39.58	32.65	4.49	24.10	30.77	23.53	33.33	28.07	14.29	25.00	40.00	5.88	25.09		
	Few-Shot	22.45	37.50	38.78	3.75	31.33	30.77	26.47	19.05	19.30	7.94	25.00	60.00	17.65	26.15		
Gemini 2.5 Pro	Zero-Shot	6.12	10.42	12.24	3.37	22.89	30.77	23.53	19.05	7.02	6.35	25.00	40.00	5.88	16.36		
	Few-Shot	8.16	8.33	10.20	2.25	21.69	30.77	17.65	23.81	15.79	7.94	25.00	20.00	5.88	15.19		
GPT-5	Zero-Shot	8.16	20.83	14.29	13.86	21.69	15.38	20.59	19.05	12.28	12.70	8.33	40.00	5.88	16.39		
	Few-Shot	6.12	10.42	18.37	5.99	14.46	0.00	17.65	9.52	21.05	14.29	16.67	20.00	23.53	13.70		
Grok-4	Zero-Shot	18.37	16.67	22.45	23.22	16.87	7.69	20.59	19.05	12.28	12.70	33.33	20.00	5.88	17.62		
	Few-Shot	14.29	22.92	20.41	8.99	20.48	15.38	26.47	19.05	7.02	7.94	16.67	20.00	17.65	16.71		
Llama 4 Maverick 17B	Zero-Shot	22.45	47.92	42.86	1.50	32.53	15.38	20.59	23.81	15.79	20.63	25.00	40.00	5.88	24.18		
	Few-Shot	20.41	39.58	46.94	3.00	31.33	15.38	23.53	14.29	28.07	19.05	33.33	60.00	17.65	27.12		
Llama 3.1 8B	Zero-Shot	69.39	77.08	65.31	20.60	54.22	38.46	50.00	47.62	42.11	31.75	41.67	80.00	17.65	48.91		
	Few-Shot	67.35	72.92	73.47	2.25	50.60	30.77	44.12	47.62	35.09	25.40	25.00	60.00	23.53	42.93		
GPT-OSS-120B	Zero-Shot	59.18	85.42	65.31	50.19	32.53	15.38	35.29	28.57	29.82	22.22	41.67	40.00	17.65	40.25		
	Few-Shot	65.31	83.33	65.31	20.22	42.17	23.08	38.24	28.57	35.09	20.63	25.00	40.00	11.76	38.36		
Qwen3 235B	Zero-Shot	26.53	52.08	48.98	12.73	24.10	15.38	23.53	28.57	22.81	12.70	25.00	60.00	17.65	28.47		
	Few-Shot	30.61	50.00	59.18	3.37	30.12	23.08	32.35	23.81	22.81	12.70	33.33	40.00	5.88	28.25		

Table 9 Abstention (refusal) rates of LLMs on our Islamic Legal Benchmark across different task complexities and prompting strategies (Pass@1 rates). Values represent abstention percentages. Lowest results per task are highlighted in **bold**

Model	Prompt	Low Complexity					Moderate					High Complexity					Overall	
		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13				
Claude Sonnet 4.5	Zero-Shot	0.00	0.00	0.00	5.62	14.46	0.00	2.94	0.00	0.00	3.17	8.33	0.00	0.00	0.00	2.66		
	Few-Shot	0.00	0.00	0.00	0.00	26.51	0.00	0.00	9.52	0.00	3.17	8.33	0.00	0.00	0.00	3.66		
DeepSeek R1	Zero-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.59	0.00	0.00	0.00	0.00	0.12		
	Few-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.33	0.00	0.00	0.00	0.64		
Gemini 2.5 Pro	Zero-Shot	0.00	0.00	0.00	0.00	6.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.46		
	Few-Shot	0.00	0.00	0.00	0.00	6.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.46		
GPT-5	Zero-Shot	0.00	0.00	0.00	0.00	6.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.46		
	Few-Shot	0.00	0.00	0.00	0.00	1.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09		
Grok-4	Zero-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		
	Few-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.59	8.33	0.00	0.00	0.00	0.76		
Llama 4 Maverick 17B	Zero-Shot	0.00	0.00	0.00	0.00	1.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09		
	Few-Shot	0.00	0.00	0.00	0.00	1.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09		
Llama 3.1 8B	Zero-Shot	0.00	0.00	4.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.31		
	Few-Shot	2.04	0.00	0.00	0.00	0.00	0.00	2.94	4.76	0.00	0.00	0.00	0.00	0.00	0.00	0.75		
GPT-OSS-120B	Zero-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		
	Few-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.67	1.28		
Qwen3 235B	Zero-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.33	0.00	0.00	0.00	0.64		
	Few-Shot	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		

phantically accepting user misinformation. Table 10 and Fig. 6 present comprehensive response distributions.

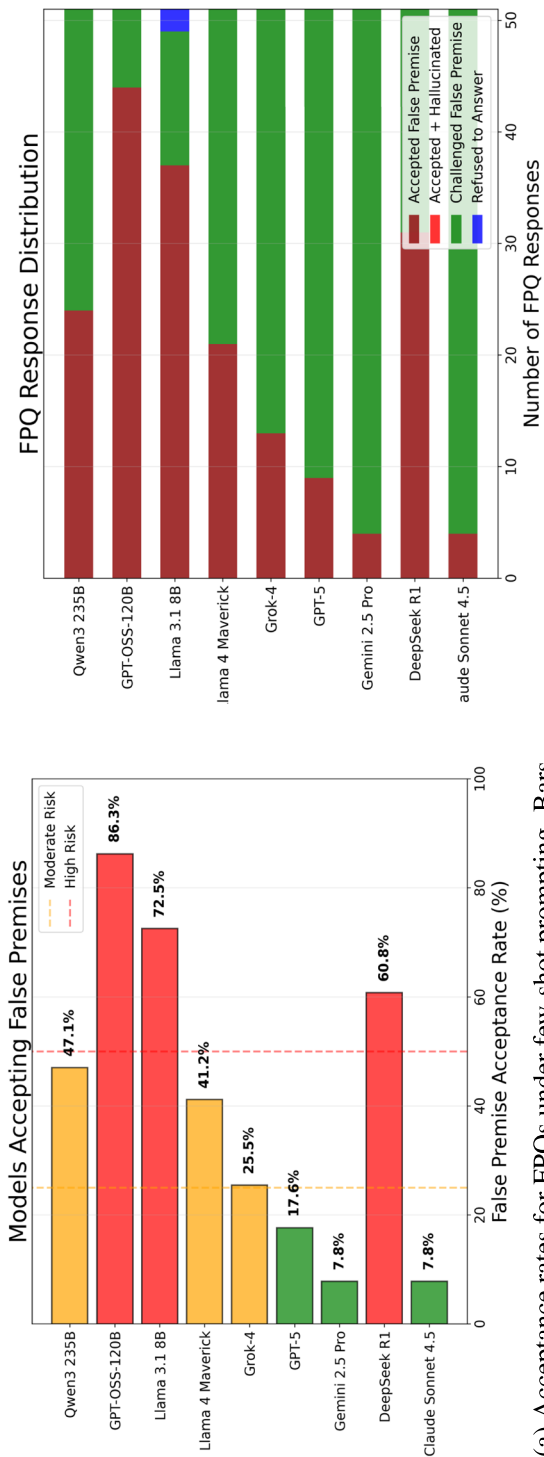
The results show sycophantic vulnerability across most evaluated models, stratifying into three distinct profiles. Claude Sonnet 4.5 and Gemini 2.5 Pro demonstrate exceptional critical reasoning by challenging 92% of false premises while accepting fewer than 8%, indicating deployment-appropriate skepticism toward user-provided misinformation. A middle tier (GPT-5, Grok-4) accepts 18-25% of false premises, substantially better than weaker systems, but still indicating that roughly one-in-four misleading assumptions pass unchallenged. The bottom tier exhibits a much higher degree of vulnerability: GPT-OSS-120B accepts 86% of false premises, with Llama 3.1 8B, DeepSeek R1, and Llama 4 Maverick showing similarly limited critical evaluation of query assumptions.

This failure appears to represent a compounding safety vulnerability beyond mere incorrectness. Models with less extensive Islamic jurisprudence grounding not only produce more incorrect answers but also appear less able to recognize and reject false information embedded in user queries. The strong negative correlation between FPQ acceptance and overall performance (Pearson's $r = -0.91$, $n = 9$, $p < 0.01$) is consistent with this pattern: weaker models may be doubly compromised, generating more errors while being less capable of detecting them, a profile that raises concerns for responsible deployment in religious guidance contexts. A rank-based robustness check yields a consistent association (Spearman's $\rho = -0.86$, $n = 9$).

Notably, in-context learning (ICL) does not appear to mitigate sycophantic vulnerability in this setting, and in some cases may exacerbate it. Average FPQ acceptance

Table 10 False Premise Query (FPQ) Analysis: Sycophancy Detection in LLM Responses. Models are evaluated on their ability to identify and reject misleading queries in Islamic law. Lower acceptance rates indicate better critical reasoning

Model	Prompt	Response Distribution			Key Metrics	
		Accepted	Challenged	Refused	Accept. Rate %	Challenge Rate %
Claude Sonnet 4.5	Zero-Shot	3	48	0	5.88	94.12
	Few-Shot	4	47	0	7.84	92.16
DeepSeek R1	Zero-Shot	26	25	0	50.98	49.02
	Few-Shot	31	20	0	60.78	39.22
Gemini 2.5 Pro	Zero-Shot	5	46	0	9.80	90.20
	Few-Shot	4	47	0	7.84	92.16
GPT-5	Zero-Shot	9	42	0	17.65	82.35
	Few-Shot	9	42	0	17.65	82.35
Grok-4	Zero-Shot	6	45	0	11.76	88.24
	Few-Shot	13	38	0	25.49	74.51
Llama 4 Maverick	Zero-Shot	23	28	0	45.10	54.90
	Few-Shot	21	30	0	41.18	58.82
Llama 3.1 8B	Zero-Shot	38	11	2	74.51	21.57
	Few-Shot	37	12	2	72.55	23.53
GPT-OSS-120B	Zero-Shot	41	10	0	80.39	19.61
	Few-Shot	44	7	0	86.27	13.73
Qwen3 235B	Zero-Shot	20	31	0	39.22	60.78
	Few-Shot	24	27	0	47.06	52.94



(a) Acceptance rates for FPQs under few-shot prompting. Bars show the proportion of misleading queries each model accepts, with green (<25%), orange (25–50%), and red (>50%) indicating increasing risk.

Fig. 6 False Premise Query (FPQ) Analysis. Few-shot prompting reveals differences across frontier models in how false premises are handled: some models tend to challenge false premises more consistently, while others accept them at higher rates, occasionally with additional hallucinated justification. These patterns may reflect varying levels of critical-reasoning robustness across architectures and prompting regimes (acceptance rates in Panel (a); response-type distributions in Panel (b))

increased from 37.25% (zero-shot) to 40.74% (few-shot), with DeepSeek R1 and GPT-OSS-120B showing more substantial increases. This pattern is somewhat counterintuitive and may suggest that few-shot examples reinforce superficial pattern-matching, increasing premise acceptance when exact knowledge is absent. Other factors, such as prompt composition or example selection, may also contribute to this effect, and we view this as a notable finding worth further exploration in future works.

6.5 False confidence: The danger zone of miscalibrated models

Figure 7 maps the critical relationship between correctness and hallucination, revealing patterns of confident incorrectness that present acute deployment risks.

Four models occupy the danger zone, characterized by low correctness coupled with high hallucination. GPT-OSS-120B and Llama 3.1 8B exhibit a high level of miscalibrated confidence, producing incorrect Islamic legal guidance in more than two-thirds of responses while hallucinating in nearly 60% of outputs. These models do not merely fall short of answering correctly; they confidently fabricate Islamic legal content, creating particularly high-risk failure modes. Llama 4 Maverick and Qwen3 235B also fall within the danger zone, nearer its boundary.

This pattern of confident incorrectness raises risks for Islamic advisory applications. Users without Islamic legal expertise to independently evaluate responses may place trust in authoritative-sounding outputs that are not always accurate, potentially leading to questionable religious practice. The limited presence of uncertainty signals (abstention rates below 1%) adds to this concern: models do not consistently indicate that their responses may be unreliable, which could encourage uncritical acceptance of potentially incorrect guidance.

Conversely, leading models demonstrate substantially better calibration. GPT-5 (67.65% correct, 21.25% hallucination) and Claude Sonnet 4.5 (65.63% correct, 19.18% hallucination) maintain roughly 3:1 ratios of correct to hallucinated responses, indicating more appropriate confidence. However, even these top performers hallucinate in approximately one-fifth of outputs, underscoring that no evaluated model achieves deployment-ready reliability without human expert oversight. Gemini 2.5 Pro and Grok-4 occupy intermediate positions, maintaining positive but less favorable correctness-to-hallucination ratios. Taken as a whole, even these ratios fall short of what high-stakes Islamic legal advisory contexts would reasonably require, where response accuracy carries direct religious consequences.

6.6 In-context learning failure: Why few-shot prompting cannot fix fundamental knowledge gaps

A central finding concerns the limited effectiveness of in-context learning (ICL) for Islamic jurisprudence reasoning. Figure 8 presents a comprehensive analysis comparing zero-shot and few-shot performance, revealing systematic ICL ineffectiveness.

The results suggest limited and inconsistent ICL effectiveness overall. Per-model changes (few-shot minus zero-shot, from the Overall column) range from -3.64 to +4.16 points. Four models improve by more than one point (Llama 3.1 8B +4.16,

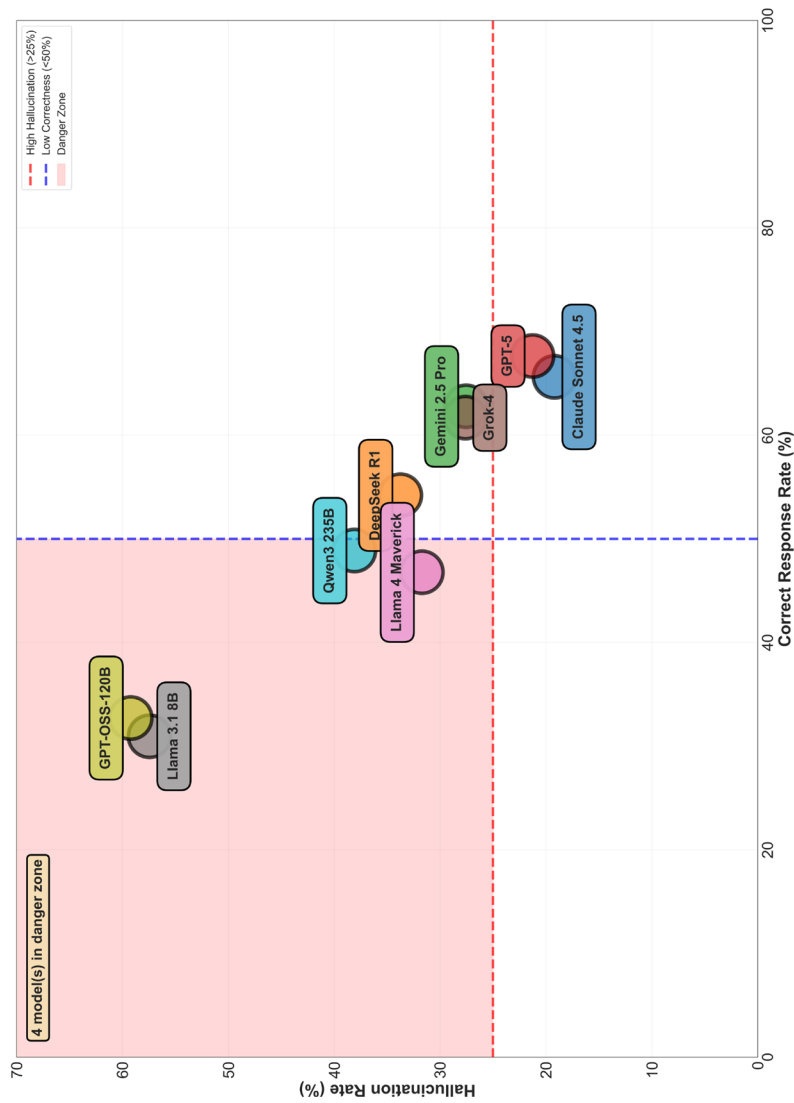
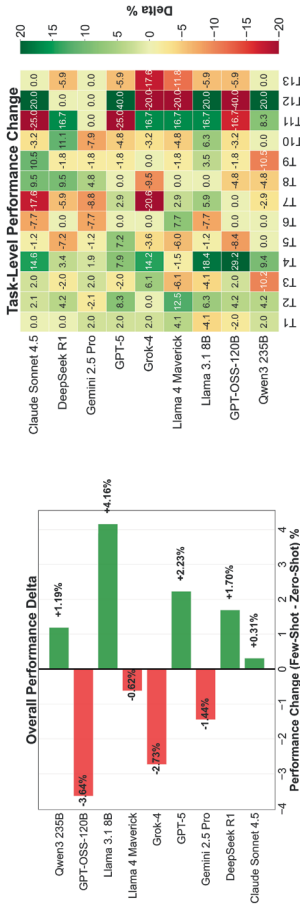
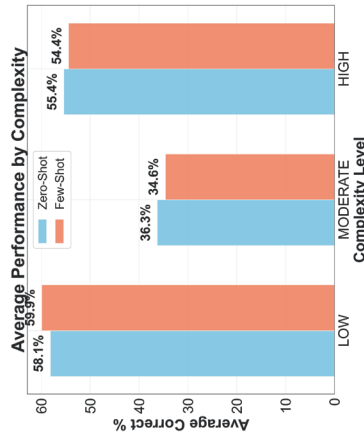


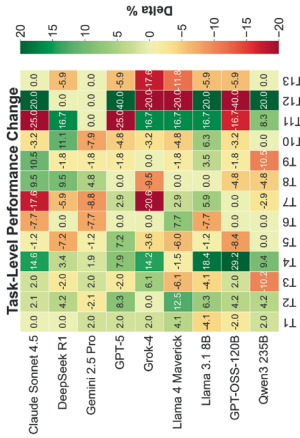
Fig. 7 Scatter plot mapping correctness versus hallucination rates under few-shot prompting. Each point represents one model, with annotations identifying specific systems. The red-shaded "danger zone" (correctness <50%, hallucination >25%) identifies models exhibiting confident incorrectness. Four models occupy this critical region, producing incorrect Islamic legal guidance while showing no uncertainty signals, creating acute safety risks for advisory applications



(a) Overall performance delta (= few-shot - zero-shot), showing minimal aggregate benefit with only two models exceeding meaningful improvement ($> 2\%$).

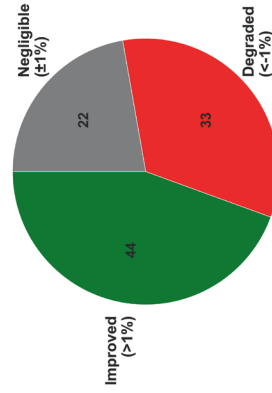


(c) Complexity-wise comparison showing negligible average benefit across Low, Moderate, and High complexity categories.



(b) Task-level delta heatmap revealing inconsistent ICL effects across tasks, with improvements on some tasks offset by degradations on others.

ICL Effectiveness Distribution



(d) Effectiveness distribution: two models fall within ± 1 point, four improve by more than a point, and three decline, indicating inconsistent and limited ICL benefit.

Fig. 8 In-Context Learning (ICL) Effectiveness Analysis. Few-shot prompting provides negligible benefit for Islamic legal reasoning: improvements are sparse, task-dependent, and offset by frequent degradations. Panels (a)–(d) present model-level gains, task variability, complexity sensitivity, and effect distributions, respectively

GPT-5 +2.23, DeepSeek R1 +1.70, Qwen3 235B +1.19), two change by less than a point (Claude Sonnet 4.5 +0.31, Llama 4 Maverick -0.62), and three decline (Gemini 2.5 Pro -1.44, Grok-4 -2.73, GPT-OSS-120B -3.64). The gains concentrate in already-weak models (Llama 3.1 8B remains at only 30.96% overall despite the largest improvement), while the strongest systems change by at most +2.23 points. These patterns suggest that few-shot examples provide limited systematic benefit for Islamic jurisprudence reasoning.

Task-level analysis offers some insight into the mechanisms underlying limited ICL benefit. The delta heatmap reveals notable performance variance across tasks for individual models: systems showing improvement on some tasks simultaneously exhibit degradation on others. For instance, Claude Sonnet 4.5 improves on T2 and T4 while degrading on T7 and T11 by a notable margin. This inconsistency suggests that few-shot examples may enable task-specific pattern matching rather than systematic reasoning enhancement, which raises some questions about the reliability of ICL-based approaches for Islamic legal reasoning.

Complexity-wise analysis suggests that ICL benefits do not consistently scale with task difficulty. Average improvements remain minimal across complexity levels (Low: +1.65%, Moderate: -0.89%, High: -1.05%), with moderate and high-complexity tasks showing slight average degradation. One possible interpretation of this pattern is that moderate-complexity tasks tend to require precise knowledge from Islamic legal texts. These tasks often demand accurate recall of specific conditions, precise doctrinal distinctions, or faithful synthesis of statutory articles. A small number of examples may be insufficient to convey knowledge that models did not acquire during earlier phases of training. Conversely, high-complexity tasks (T10-T13: ‘illah identification, analogical reasoning, cross-madhab synthesis, maxim mapping) allow for more flexible reasoning and verbose explanation, which may allow models to draw on general reasoning capabilities without requiring memorized legal provisions or committing to a specific position. Models can often discuss legal principles discursively without necessarily knowing the precise textual requirements.

The FPQ degradation pattern offers further evidence relevant to ICL limitations. As noted in Section 6.4, average FPQ acceptance increased from 37.25% (zero-shot) to 40.74% (few-shot), a 3.49 percentage point increase. This counterintuitive result may suggest that few-shot examples could reinforce sycophantic tendencies in some cases, potentially encouraging models to accept user premises rather than critically evaluate them.

Taken together, these findings suggest that in-context learning may provide limited reasoning enhancement for Islamic jurisprudence tasks in this setting. One plausible contributing factor is the limited presence of foundational Islamic legal knowledge in pre-training corpora, which few-shot examples may not fully compensate for. Models with less grounding in Islamic law, Hadith traditions, madhab methodologies, and jurisprudential principles may rely more heavily on surface-level pattern matching when presented with few-shot examples. This tendency appears more pronounced for moderate-complexity tasks (T7-T9) that require precise recall of legal conditions and statutory provisions, while high-complexity tasks (T10-T13) that reward discursive reasoning over precision may be comparatively less affected, though still showing some limitations.

This pattern may also relate to the way LLMs are generally optimized for semantically coherent text continuation rather than exact factual retrieval. Their distributed, meaning-based internal representations could lead to plausible but occasionally fabricated completions through semantic interpolation, particularly when tasks require structured enumeration of conditions. In contrast, high-complexity juridical reasoning tasks may activate general logical patterns learned during post-training (SFT, RLHF), which could allow models to approximate competence on ‘illah identification and maxim mapping even without precise textual knowledge. While we offer this as one possible explanation among others, this result may carry practical implications: prompt engineering alone may not fully substitute for gaps in training data. Continued efforts to develop models trained on more comprehensive corpora spanning classical and contemporary Islamic legal scholarship (8th-20th century), including Hadith collections, madhab-specific jurisprudential works, codified legal compendia, and contemporary databases, could help address these gaps. Such efforts may contribute to more reliable Islamic legal reasoning and advisory capabilities in future systems.

6.7 Closed-source versus open-source: Quantifying the performance gap

Figure 9 quantifies the substantial performance gap between closed-source and open-source models across three key metrics: correctness, hallucination, and abstention.

Closed-source models and open-source models differ by 21.73 percentage points in correctness (few-shot means of 64.44% and 42.71%, respectively). This substantial gap persists across the performance distribution: all four closed-source models exceed the open-source average, with even the lowest-performing closed-source system (Grok-4) outperforming all five open-source alternatives. Only DeepSeek R1 approaches competitive performance among open-source models.

Hallucination analysis reveals notable additional gaps. Closed-source models show an average of 23.89% hallucination compared to 44.04% for open-source systems, a 20.15 percentage point gap. This disparity appears most pronounced at the extremes: while the best closed-source model (Claude) and best open-source model (DeepSeek R1) both maintain sub-40% rates, the lowest performing open-source models (GPT-OSS-120B, Llama 3.1 8B) show hallucination exceeding 50%. Abstention is negligible for both groups, with closed-source models abstaining marginally more than open-source systems (1.24% versus 0.55% on average); neither tier shows a meaningful uncertainty-calibration advantage. Given open-source models’ substantially higher hallucination rates, their near-total absence of abstention is the more concerning: more proportionate calibration would involve higher abstention rates for models achieving only 30-55% correctness while hallucinating at 30-60% rates.

These gaps likely reflect multiple factors: (1) larger model scale and greater training compute for closed-source systems, (2) more extensive safety fine-tuning and reinforcement learning from human feedback procedures, (3) proprietary training data potentially including Islamic legal texts absent from open-source training sets, and (4) more sophisticated inference-time techniques including chain-of-thought prompting, self-consistency sampling, and answer verification. The current performance gap creates concerning access inequities: only well-resourced organizations

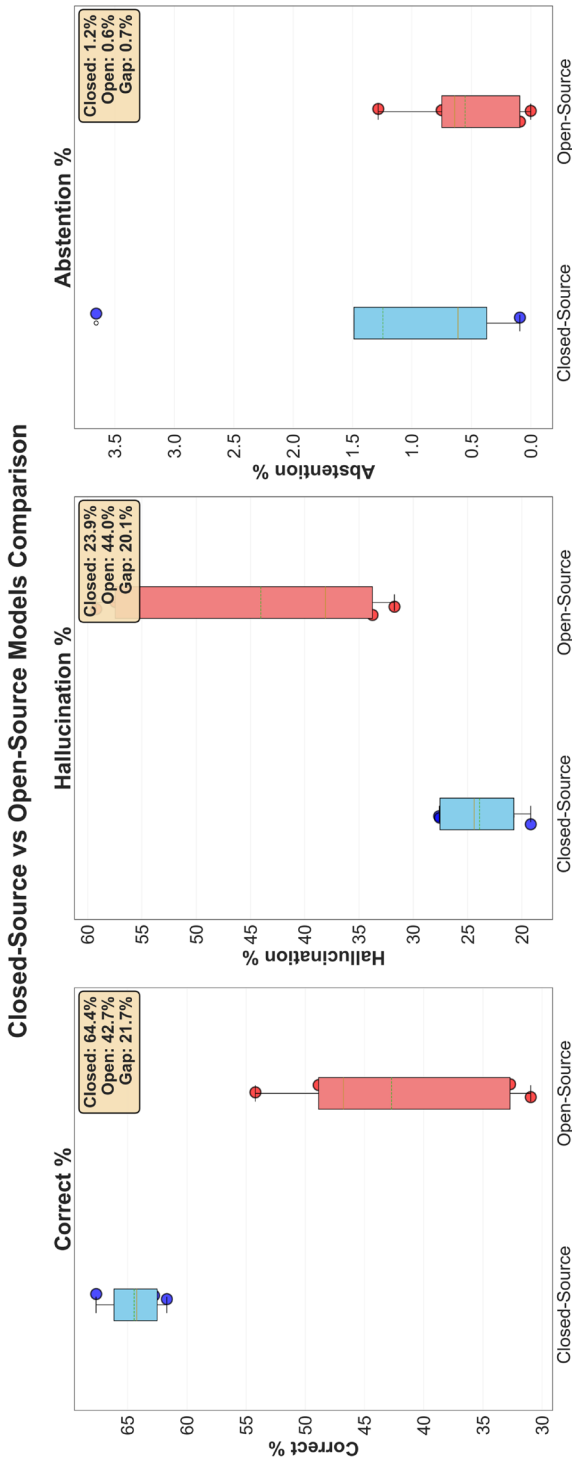


Fig. 9 Comparative analysis of closed-source versus open-source model performance under few-shot prompting across correctness, hallucination, and abstention metrics. Box plots display median (solid line), mean (dashed line), and individual model performance (scatter points). Closed-source models (blue) substantially outperform open-source models (red) on correctness while maintaining lower hallucination rates. The 21.73 percentage point correctness gap and 20.15 percentage point hallucination gap underscore substantial advantages of proprietary systems for Islamic jurisprudence reasoning

capable of affording closed-source API access can deploy models with acceptable Islamic jurisprudence reasoning capabilities, while open-source alternatives exhibit inadequate performance profiles.

7 Discussion

7.1 The mid-complexity failure zone

Our results reveal a paradoxical non-monotonic performance **Mid-Complexity Failure Zone** pattern: LLM accuracy does not change monotonically with task complexity. Instead, under few-shot prompting we observe elevated correct answer rates on low-complexity tasks (e.g., Task 4: 79.03–97.75% across models), a drop on moderate-complexity tasks (e.g., Task 7: 8.82–47.06% across models), and paradoxically improved performance on high-complexity tasks (e.g., Task 13: 47.06–82.35% across models). Even GPT-5, the top-performing model, exemplifies this pattern: achieving 76.49% average correctness on low-complexity tasks, dropping to 54.03% on moderate-complexity tasks, and recovering to 64.59% on high-complexity tasks. There is a clear U-shaped performance trough where moderate-complexity performance falls 22 percentage points below low-complexity performance despite the model’s overall superiority. Figure 10 illustrates this phenomenon. It is important to note that this non-monotonic pattern may partly reflect how complexity was operationalized, since our complexity tiers were defined based on cognitive demands placed on human scholars rather than machine difficulty, and thus, the boundaries between tiers may not perfectly correspond to distinct computational challenges for LLMs.

A structurally similar pattern has been described in human–robot interaction research as the “*uncanny valley*”, where increasing human-likeness generally improves user comfort *until a sudden non-monotonic dip* in the middle, after which trust rises again at full realism (Mori et al. 2012). The danger of this middle trough lies in its *misleading plausibility*: the agent appears convincingly human while still behaving in unreliable or unsettling ways. By analogy, our findings suggest a comparable *reasoning uncanny valley*, in which LLMs seem most competent at moderate complexity precisely when they are most likely to be incorrect, creating a deceptively high-risk failure zone.

This Mid-Complexity Failure Zone carries significant practical risks. Moderate-complexity responses appear authoritative, employ correct terminology, cite plausible legal frameworks, and demonstrate apparent sophistication. Unlike low-complexity queries, where factual errors may be readily detected, or high-complexity queries recognized as speculative, moderate-complexity responses present precise-seeming answers to questions with objectively correct answers grounded in specific texts.

Legal Dual-Process Theories and Cognitive Misalignment. This failure pattern finds theoretical grounding in legal dual-process theories, which propose that legal reasoning emerges from two distinct cognitive systems (Ronkainen 2011). *Type 1 processing* operates through fast, intuitive, gist-based reasoning and narrative pattern recognition. *Type 2 processing* enables slow, deliberate, detail-sensitive doctrinal application requiring careful textual analysis. Legal scholars have argued that tasks

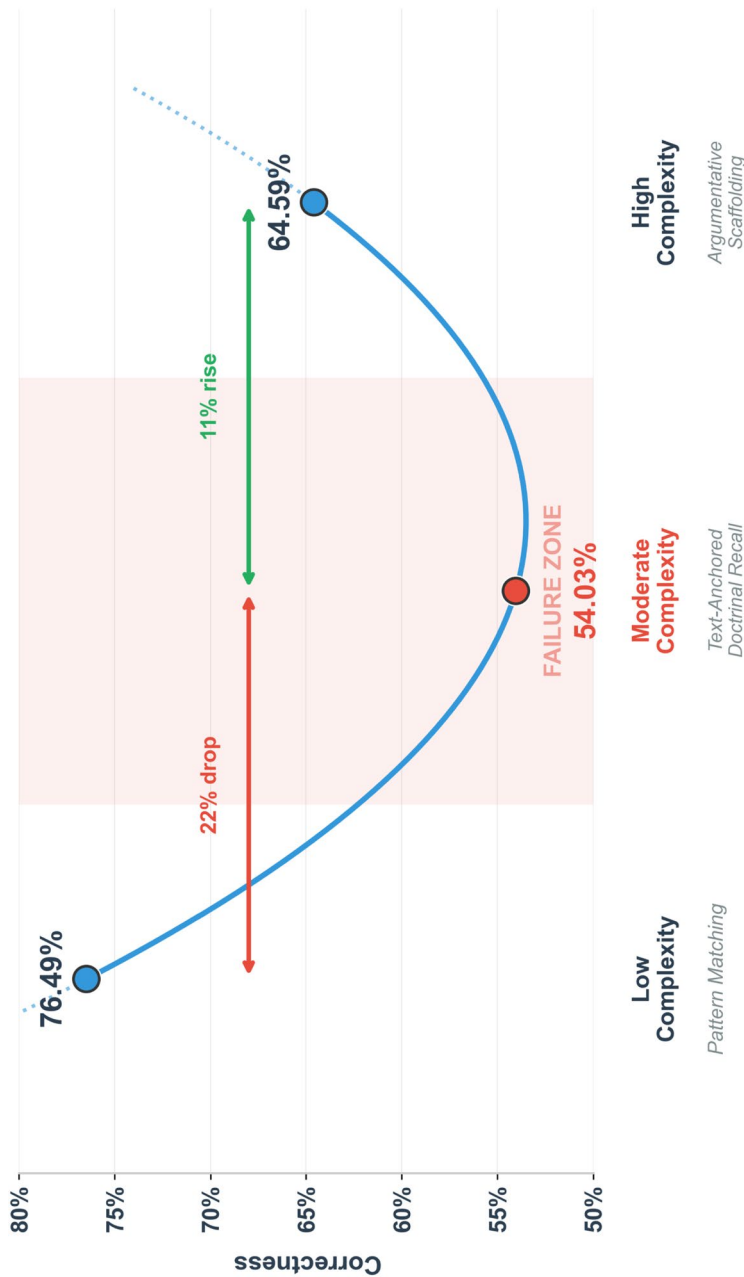


Fig. 10 The Mid-Complexity Failure Zone: Performance pattern for GPT-5, the top-performing model, across complexity tiers under few-shot prompting. The model exhibits a clear performance trough: 76.49% on low-complexity tasks (pattern matching), plummeting to 54.03% on moderate-complexity tasks requiring specific text-anchored doctrinal recall (the failure zone), and recovering to 64.59% on high-complexity tasks relying on generalized argumentative scaffolding. The shaded region highlights the dangerous zone where even state-of-the-art models regularly fail yet produce authoritative-seeming outputs

requiring simultaneous engagement of both systems, which recall specific textual rules (Type 2) while constructing structured legal arguments (Type 1), pose particular cognitive challenges and may increase the likelihood of errors (Ronkainen 2011). Our mid-complexity tasks occupy precisely this zone, requiring text-anchored doctrinal recall alongside coherent reasoning frameworks, creating conditions under which performance reliably degrades.

The mechanisms underlying this failure zone illuminate fundamental architectural characteristics of LLMs and suggest why human complexity classifications may fail to predict machine difficulty. Our complexity classifications were assigned by Islamic law experts based on the cognitive demands these tasks place on **human scholars** trained in traditional jurisprudential methodology. However, this human-centric difficulty hierarchy inverts for LLMs due to fundamental architectural differences in information processing. This cognitive gap connects to Kahneman's dual-process model of reasoning (Kahneman 2011), which distinguishes between *System 1* (fast, automatic, pattern-based processing) and *System 2* (slow, deliberate, analytical reasoning). This framework has been extended to deep learning contexts by Bengio (Bengio 2022).

Islamic legal reasoning (*ijtihad*) is grounded in a disciplined methodology that integrates textual interpretation, evaluation of evidentiary hierarchies, and rule-governed argumentation to derive rulings from the Qur'an, Sunnah, consensus, and analogical reasoning (Hallaq 1997). Human scholars therefore approach this process as a quintessential *System 2* task, demanding slow, effortful, and principled deliberation rather than intuitive or heuristic judgment (Kahneman 2011). LLMs, however, operate entirely as an amplified form of *System 1*: large-scale statistical pattern recognition running through opaque architectures. LLMs cannot verify factual claims or recognize knowledge gaps for sources never encountered in training data. Instead, they approximate reasoning through verbose pattern generation, producing outputs that *linguistically resemble* analytical thought without actual grounding. This creates a dangerous mismatch: moderate-complexity Islamic legal tasks demand both Type 1 reasoning (constructing coherent juridical arguments) and Type 2 precision (recalling exact textual specifications), yet LLMs can only provide the former through statistical patterns while fabricating the latter through semantic interpolation.

This also raises a broader concern: users without legal training lack the evaluative capacity to distinguish linguistically polished output from substantively accurate reasoning. Such users are therefore especially vulnerable to being misled by fluent but incorrect responses—a risk noted in prior legal-AI evaluations (Schweitzer and Conrads 2025). A parallel issue was observed in software communities: StackOverflow prohibited generative-AI answers in 2022 because their apparent “quality” frequently concealed errors (Stack Overflow Meta Community 2022).

This architectural difference creates systematic performance inversions across complexity tiers. Low-complexity tasks require only surface-level pattern matching (79.03–97.75% accuracy). High-complexity tasks ('illah identification, maxim mapping, cross-school synthesis) perform comparatively better (47.06–82.35%) because models rely on generalized argumentative scaffolding and can maintain abstract reasoning without committing to precise doctrinal facts. Moderate-complexity tasks occupy the failure zone because they demand what LLMs lack: specific text-

anchored doctrinal recall. Tasks requiring condition enumeration (T7), comparative distinctions (T8), or statute-like synthesis (T9) demand factual precision with single correct answers determinable only through direct textual access. LLMs instead produce *semantic interpolation*, plausible completions maintaining discourse coherence without factual grounding. This structural limitation produces elevated hallucination rates.

One example of Claude Sonnet 4.5's performance on *Mukhtaṣar al-Qudūrī* illustrates this paradox (Fig. 11). On moderate-complexity T7 (enumerating *salām* contract conditions), Claude **failed**, fabricating "availability of commodity" and "specification of place of delivery". These are absent from Abū Ḥanīfa's actual six-condition formulation. Yet on high-complexity T10 from the *same book* (identifying 'illah for debt-for-debt prohibition), Claude **succeeded** completely, providing correct analysis of *ribā al-nasī'ah*, *gharar*, and possession requirements without hallucination. The difference lies in task structure: abstract legal *principles* activate semantic reasoning circuits leveraging deeply learned patterns; enumerating specific *conditions* triggers semantic interpolation yielding fabricated yet authoritative-sounding content. This pattern replicates systematically across approximately 9,605 instances.³ A single model might fail on enumerating *salam* contract conditions (T7), while succeeding on multiple high-complexity queries involving *salam* contracts (T10-T13), generating multiple documented paradox instances from the same underlying knowledge gap for all 9 LLMs: models failed moderate-complexity verbatim retrieval (58-97% failure rates) yet succeeded on thematically similar high-complexity reasoning from identical source texts (38-70% success rates), consistent with verbose reasoning masking gaps in accessible foundational knowledge.

The Illusion of Knowledge and Dangerous Misplaced Confidence. This failure mode aligns with established cognitive biases. The *Dunning-Kruger effect* (Kruger and Dunning 2000) demonstrates that individuals with low competence systematically overestimate their abilities, lacking metacognitive capacity to recognize incompetence. Similarly, Kahneman's *Overconfidence effect* (Kahneman 2011; Dobelli 2013) describes how individuals overestimate judgment accuracy when operating with incomplete information. We observe a structurally analogous *Illusion of Knowledge* in LLMs: models generate confident, technically sophisticated responses about Islamic legal conditions never encountered in training data, exhibiting what Dobelli terms *chauffeur knowledge* (Dobelli 2013), surface-level fluency mimicking expertise without underlying comprehension. Like a chauffeur who memorizes a physicist's lecture but cannot answer deeper questions, LLMs produce linguistically sophisticated outputs that create an appearance of expertise without factual grounding.

This manifests empirically through near-zero abstention rates (<1%) despite 30-60% hallucination rates. A model confidently enumerating seven *nikāḥ* conditions when texts specify five appears knowledgeable while fundamentally misrepresenting Islamic legal requirements. Our FPQ results (Section 6.4) reinforce this pattern: models accept 40-86% of deliberately incorrect premises, building elaborate justifications rather than critically evaluating validity. When asked to analyze the "major Maliki

³ Identified from 12,924 total model responses: 718 items × 9 models × 2 prompting conditions (zero-shot, few-shot).

Cognitive Misalignment Example - Claude Sonnet 4.5

Task: Condition Enumeration Book: Mukhtasar al-Quduri Score: 1/2 (Partially Correct), Hallucination: TRUE Query: "List the specific conditions (shurut) for the validity of a Salam contract (forward contract) according to the text, as held by Abu Hanifa." Ground Truth (6 specific conditions): 1. Known type (جنس معلوم) 2. Known variety (نوع معلوم) 3. Known description (وصفة معلومة) 4. Known quantity (مقدار معلوم) 5. Known due date (أجل معلوم) 6. Knowledge of the capital (مقدار رأس المال) **Plus**: Capital must be taken before parties separate, and neither capital nor subject can be disposed of before possession. Model Response (hallucinated conditions 5 & 6): 1. Full payment of the price in contracting session ✓ 2. Precise specification of the commodity ✓ 3. Determination of quantity ✓ 4. Specification of delivery term ✓ 5. **Availability of the commodity** ✗ **HALLUCINATED** - Not in ground truth 6. **Specification of place of delivery** ✗ **HALLUCINATED** - Not in ground truth Why it failed: Model added two conditions (availability, place) not in the ground truth, demonstrating it lacks verbatim knowledge and fabricates plausible-sounding requirements.	HIGH TASK (T10) Task: Illah identification and doctrinal rationale Book: Mukhtasar al-Quduri (same book) Score: 2/2 (Correct), Hallucination: FALSE Query: "The text discusses the permissibility of selling a debt (دين) for a debt (دين) and the conditions under which it is allowed or prohibited. What is the underlying rationale for the general prohibition of selling a debt for a debt, and what specific condition, if met, allows for its permissibility according to the Hanafi school?" Ground Truth: The rationale is to prevent **riba al-nasi'ah** (usury of delay) and **gharar** (uncertainty). Permissible if debt is exchanged for a tangible, present item (عين حاضر) immediately taken into possession. Model Response: **CORRECT** # Chain-of-Thought Analysis: Bay al-Dayn bi al-Dayn in Hanafi Fiqh ## Step 1: Identifying the General Prohibition According to al-Quduri in "Mukhtasar al-Quduri" ... Step 2: Underlying Rationales (Illah) for the Prohibition The Hanafi school identifies multiple interconnected rationales in this prohibition: 1. **Gharar (Excessive Uncertainty)**: ... 2. **Absence of Immediate Possession (Qabz)**: ... 3. **Gateway to Riba**: ... 4. **Instability of Obligations**: ... ## Step 3: The Specific Permissibility Condition ... Provided detailed Chain-of-Thought analysis: - Identified gharar, absence of possession, gateway to riba as rationales ✓ - Explained that possession (qabz) of at least one counter-value in contracting session makes it permissible ✓ - Connected to Hanafi principles correctly ✓ - No hallucinations, grounded in source text ✓ Why it succeeded: Task allows verbose reasoning about "principles" (gharar, qabz, riba) without requiring memorization of exact conditions and it engaged the model's semantic reasoning capabilities rather than exact conditions recall. Model can discuss legal rationales abstractly and appear competent.
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Fig. 11 Cognitive misalignment illustrated: Claude Sonnet 4.5 performance on two tasks from *Mukhtasar al-Quduri*. (Left) Moderate-complexity T7 (condition enumeration for *salam* contracts) results in failure with two fabricated conditions absent from Abū Hanifa's actual six-condition formulation. (Right) High-complexity T10 (identifying *illah* for debt-for-debt prohibition) from the same book succeeds completely with correct analysis of *riba al-nasi'ah*, *gharar*, and possession requirements

work *Badā' al-Ṣanā'i'* (actually Hanafi), weaker models generate Maliki analyses, sycophantically accepting false premises and generating responses consistent with query framing rather than retrieving contradictory knowledge.

Implications. The ICL failure (Section 6.6) amplifies this pattern: few-shot prompting provides negligible and inconsistent benefit (per-model changes span -3.64 to +4.16 points, with gains concentrated in already-weak models) and worsens FPQ acceptance by 3.49 percentage points, reinforcing sycophantic behaviors rather than enhancing critical reasoning. The failure zone cannot be traversed through prompting. Few-shot examples cannot teach content from classical fiqh texts never encountered during pre-training. First, aggregate performance metrics obscure critical failure modes. A model achieving 55% overall correctness appears marginally useful, yet if performance concentrates on trivial queries and verbose speculation while failing on precise-knowledge tasks, the metric severely misrepresents deployment readiness. Second, the most high-risk deployment scenarios involve precisely those moderate-complexity queries where users receive authoritative-seeming answers grounded in hallucination rather than authentic knowledge. Third, overcoming the mid-complexity failure zone requires comprehensive domain-specific pre-training on classical Hadith collections, *madhab*-specific jurisprudential works, and codified Islamic legal works—not post-hoc prompting or instruction tuning interventions.

7.2 Alignment with broader LLM reasoning limitations

The “mid-complexity failure zone” we observe validates established research that LLMs prioritize surface-level pattern matching over genuine reasoning. Models’ high degree of failure on precise enumeration tasks mirrors the fragility documented by Mirzadeh et al. (2024) and Nezhurina et al. (2025), where architectural limitations lead to high-confidence confabulation on logically simple problems. This sensitivity to task structure rather than semantic complexity aligns with Roh et al. (2025) and Chatziveroglou et al. (2025), who found that minor formulation changes trigger reasoning breakdowns. However, our findings extend beyond the directional retrieval asymmetry of the Reversal Curse (Berglund et al. 2024); we identify a fundamental “correlation vacuum” where the absence of classical *fiqh* texts in pre-training forces models to rely on semantic interpolation to mask knowledge voids.

Mechanistically, this behavior is consistent with models defaulting to spurious correlations and shortcuts as described by Enström et al. (2024), Zhou et al. (2024), and Steinmann et al. (2024). The sycophancy observed in our FPQs acceptance further instantiates the entailment bias noted by Yuan et al. (2024). Crucially, while standard spurious correlations may be addressable through robustness interventions (Gui 2025; Shueh et al. 2025), the ICL failure demonstrates that post-hoc prompting cannot remedy these specific deficits. Unlike reasoning errors, the absence of domain knowledge is not a retrieval failure but a representational gap, indicating that reliability in high-stakes Islamic law requires comprehensive pre-training on authoritative sources rather than prompt engineering.

8 Limitations and future directions

Our work establishes the first comprehensive evaluation framework for Islamic legal reasoning in LLMs, opening multiple promising directions for future research and development.

Multilingual evaluation opportunities While our benchmark provides a comprehensive English-language evaluation, extending to Arabic and other languages spoken across Muslim-majority regions represents a valuable direction for future work. Given that classical Islamic legal texts were originally composed dominantly in Arabic, bilingual evaluation would enable important investigations: whether models exhibit differential Islamic legal knowledge depending on query language, whether translation introduces systematic artifacts, and whether models trained predominantly on English Islamic texts demonstrate gaps in classical Arabic legal terminology. Building on this, future work within our research agenda includes first developing an Arabic version of `IslamicLegalBench` to empirically quantify the effect of language and cultural grounding on model performance and then extending to other languages as well.

Internal knowledge & retrieval augmentation Our study offers a comprehensive LLM evaluation for Islamic legal domains through internal knowledge assessment, or **knowledge-based benchmarking**. We test models' ability to reason about Islamic law using only their pre-trained parameters and in-context few-shot learning, without external retrieval, knowledge bases, or tool augmentation. This directly addresses whether current foundation models contain sufficient Islamic legal knowledge in their training data. Our results are consistent with substantial gaps in Islamic legal knowledge in current training corpora, and indicate that prompting techniques did not compensate for these gaps under our evaluation conditions. However, since evaluation is conducted in a closed-book setting, observed failures may reflect limitations in recall or task-format sensitivity in addition to gaps in underlying domain knowledge. Furthermore, while our benchmark encompasses both memory-reliant tasks (e.g., recalling specific rulings, texts, or scholarly positions) and reasoning-reliant tasks (e.g., applying jurisprudential principles to novel scenarios), the two are deeply interdependent in Islamic legal reasoning: foundational knowledge is a prerequisite for sound jurisprudential reasoning, and reasoning competence is necessary to apply knowledge meaningfully. Our tier structure operationalizes this distinction directly, and all results are reported per tier to allow separate inspection of recall- and reasoning-oriented performance. We note that by instance count the benchmark is weighted toward recall-oriented tasks (509 of 718 instances in the Low tier, of which 267 are concept-existence checks); accordingly, claims about reasoning capability should be read as grounded primarily in the 97 High-complexity instances, and aggregate correctness should not be read as a measure of jurisprudential reasoning alone.

Contemporary LLM applications frequently employ Retrieval-Augmented Generation (RAG) (Shuster et al. 2021; Cui et al. 2023; Peng et al. 2023; Savelka et al. 2023), specialized fine-tuning (Tian et al. 2023; Razumovskaia et al. 2024; Zhang

et al. 2024), factuality-oriented decoding (Shi et al. 2024; Mallen et al. 2023; Li et al. 2023; Chuang et al. 2024), and external database checks (Chern et al. 2025; Peng et al. 2023; Qin et al. 2024; Gou et al. 2024) to augment base models with external knowledge. Evaluating such enhanced systems is part of comprehensive benchmarking and is our immediate next research direction. In Western legal contexts, Magesh et al. (2025) shows that RAG-based tools such as Lexis+ AI and Westlaw AI-Assisted Research markedly outperform base models yet still exhibit hallucination rates of 17–33%. Analogous systematic evaluation is urgently needed for Islamic legal applications that couple LLMs with retrieval over Islamic legal databases, Hadith collections, and fatwa repositories. However, specialized Islamic RAG-enabled platforms and LLMs have emerged, such as Usul.AI (Usul 2025) and Fanar (Team et al. 2025), and their performance will be evaluated in future works.

LLMs will likely need to excel at both internal knowledge and external augmentation to be safely integrated into Islamic legal advisory systems. Our results suggest that gaps in background Islamic legal knowledge may limit the reliability of LLMs as sources of religious counsel, even with advances in retrieval or tool use. Limited base-model knowledge may introduce structural vulnerabilities: models may struggle to reliably formulate retrieval queries, assess the relevance and authenticity of retrieved texts, detect contradictions between sources, or distinguish authoritative rulings from minority or superseded positions.

While retrieval-based and related methods have shown some reduction in hallucinations and improved accuracy, reliability, and faithfulness in general domains (Chen et al. 2024; Siriwardhana et al. 2023; Ram et al. 2023; Cheng et al. 2023), they face specific limitations in Islamic law. Their effectiveness hinges on retrieval quality (Wu et al. 2025) and on correctly parsing often ambiguous, vernacular user queries rather than technical legal formulations (Tonmoy et al. 2024). Retrieving precise information from extensive Islamic corpora spanning over 1,200 years, seven schools of jurisprudence, and multiple subdomains is computationally demanding and requires continual updates to incorporate contemporary and evolving interpretations (Siriwardhana et al. 2023). Knowledge bases may also contain conflicting information (Wang et al. 2025; Yu et al. 2024; Gao et al. 2024): for instance, when classical rulings are superseded by contemporary consensus, when schools hold contradictory positions on an issue, or when the authority of a source depends on its chain of transmission (*isnād*). In such cases, the retrieval module and the model must identify outdated, jurisdictionally irrelevant, or methodologically inappropriate sources and privilege binding, apposite rulings, a task that is highly dependent on robust foundational Islamic legal knowledge.

Advancing automated evaluation methodology Our LLM-as-a-Judge framework using o3 enables scalable assessment and shows strong correlation with expert annotations, yet automated evaluation in specialized legal domains remains imperfect. Additionally, two forms of prompt sensitivity remain untested here: sensitivity of the evaluated models' responses to paraphrased query formulations, and sensitivity of the LLM-as-a-Judge to its own evaluation prompt. Our inter-annotator analysis (Cohen's $\kappa = 0.73$ and 0.72 between blinded human experts, with substantial expert–judge agreement) supports the reliability of the scoring rubric as applied to the responses we collected, but does not rule out response-level variation under alternative phras-

ings. A systematic paraphrase-robustness analysis is a priority for future work. Evaluation metrics may not fully capture real-world reliability (Ji et al. 2023; Zhang et al. 2025; Lucas et al. 2023). Our focus on published classical texts provides rigorous grounding, though future work could incorporate oral traditions and contemporary interpretive developments. Our hallucination metrics offer a useful signal but may reflect dataset biases or lack sufficient comprehensiveness, nuance, or task-specificity (Kang et al. 2024), so it remains unclear whether technical advances can fully resolve hallucination. Our binary hallucination flagging (True/False) is efficient, but finer-grained taxonomies (omission vs. commission, minor vs. major distortion, plausible vs. implausible fabrication) and their differential risks for Islamic legal advisory systems warrant study. Although we validated ground truth answers with experts, jurisprudential pluralism means some questions legitimately admit multiple answers across schools, periods, and methodologies. Our framework partially accommodates such variation, providing a basis for future work on more sophisticated automated evaluation in a pluralistic domain like Islamic jurisprudence.

Model coverage and emerging systems We evaluated nine state-of-the-art LLMs representing both closed-source (GPT-5, Claude Sonnet 4.5, Gemini 2.5 Pro, Grok-4) and open-source (DeepSeek R1, Llama 4 Maverick, Llama 3.1 8B, GPT-OSS-120B, Qwen3 235B) systems available at the time of benchmarking (mid 2025 to late 2025). The rapid pace of LLM development means newer models with potentially improved Islamic legal capabilities will continue to emerge, and our benchmark provides a standardized framework for evaluating such systems as they become available. Our findings characterize current system capabilities rather than fundamental architectural limits. Moreover, our focus on general-purpose foundation models that Muslims increasingly consult for fatwas and religious rulings provides a crucial baseline assessment. Future work should systematically compare these general-purpose models against domain-specialized Arabic and Islamic AI systems such as ALLaM (Bari et al. 2024), Fanar (Team et al. 2025), and Ansari AI⁴, and against culturally aligned evaluation resources such as ILMAAM (Nacar et al. 2025), which are designed to address precisely these cultural and linguistic alignment gaps. Several of these systems (e.g., Fanar and Ansari AI) are retrieval-augmented, which differs from the closed-book protocol used here; evaluating them therefore requires the RAG-condition protocol identified above as our immediate next research direction. This includes evaluating agentic Islamic AI systems and extending IslamicLegalBench to other languages to disentangle the effects of language and cultural grounding on model performance.

These limitations notwithstanding, IslamicLegalBench provides a foundational evaluation framework for Islamic legal reasoning in LLMs, revealing critical knowledge gaps and failure modes with profound implications for Muslim communities worldwide, who are increasingly relying on AI systems for religious guidance. Our findings establish the important need for training foundation models on comprehensive Islamic legal corpora and developing rigorous evaluation methodologies before deploying such systems in sensitive religious advisory contexts.

⁴<https://askansari.ai/welcome>

9 Conclusion

As millions of Muslims increasingly consult large language models (LLMs) for religious and legal guidance, we ask a fundamental question: can AI systems trained predominantly on Western texts reliably reason about Islamic jurisprudence? Evaluating nine state-of-the-art LLMs, we find performance patterns consistent with substantial Islamic legal knowledge gaps in current foundation models, gaps that standard prompting did not close in our closed-book evaluation. We introduce *IslamicLegalBench*, the first comprehensive benchmark covering the full spectrum of Islamic legal reasoning. Built from 37 foundational texts spanning 1,200 years and seven schools of jurisprudence, it comprises 718 evaluation instances across 13 task types organized by complexity. We assess capabilities from bibliographical recall to advanced jurisprudential reasoning, including legal rationale identification (*'illah*), analogical application (*qiyās*), comparative rule distinction, and cross-school synthesis, under both zero-shot and few-shot prompting conditions.

Taken together, our results reveal severe knowledge gaps across all evaluated models. The best system reaches 67.65% correctness with 21.25% hallucination, while several models fall below 35% correctness and exceed 55% hallucination. Few-shot prompting offers no consistent benefit: per-model changes range from -3.64 to +4.16 points, with gains concentrated in already-weak models, consistent with Islamic legal content being sparsely represented in pretraining, with prompting unable to substitute for it under the conditions tested. We further identify a “Mid-Complexity Failure Zone”: moderate-complexity tasks that require specific, text-anchored doctrinal recall show the highest degree of failure, with correctness often between 8.82 and 47.06% and hallucination between 35 and 73%, whereas high-complexity reasoning tasks fare comparatively better (47.06 to 82.35%) because models rely on generalized argumentative scaffolding and can maintain abstract reasoning without committing to exact doctrinal text. This pattern yields thousands of instances where fluent argumentation masks missing jurisprudential grounding rather than demonstrating genuine competence. The safety risks are also substantial. Under few-shot prompting, five of the nine models accept misleading Islamic legal assumptions at rates above 40% (worst 86.27%), and few-shot prompting increases this behavior by 3.49 percentage points.

These findings imply that progress will not come from prompt engineering alone, but from pretraining on comprehensive Islamic legal corpora. In-Context Learning (ICL) failures show that few-shot examples cannot supply content never seen during pretraining. Our benchmark provides the first systematic evaluation framework for Islamic legal AI systems and exposes critical weaknesses in platforms that Muslims increasingly treat as sources of religious guidance. Given these documented knowledge gaps and failure modes, general-purpose LLMs are not ready for Islamic legal advisory use without substantial improvements in foundational training, rigorous expert oversight, and explicit communication of limitations. Only community-driven efforts to train specialized models, strengthen evaluation, and establish ethical deployment guidelines can ensure that Islamic legal AI serves rather than undermines the spiritual needs of Muslim communities.

A Comprehensive list of source texts

This appendix provides a comprehensive listing of all primary texts of Islamic law included in our source dataset. It includes full bibliographic details organized by school of thought, author, and historical period.

Table 11 Foundational Texts of Islamic Law Used in the Source Dataset

School	Title	Author	Period
Hanafi	Mukhtasar al-Quduri	Abū al-Ḥasan Aḥmad ibn Muḥammad al-Qudūrī	Early Classical (d. 428 H / 1037 CE)
Hanafi	Kitab al-Buyu'	Muḥammad ibn al-Ḥasan al-Shaybānī	Early Classical (d. 189 H / 805 CE)
Hanafi	Badā'i' al-Ṣanā'i' fī Tartīb al-Sharā'i'	'Alā' al-Dīn al-Kāsānī	Late Classical (d. 587 H / 1191 CE)
Hanafi	Majma' al-Ḍamānāt	Ghānim ibn Muḥammad al-Baghdādī	Late Classical / Ottoman (d. 1030 H / 1621 CE)
Hanafi	al-Islām: 'Aqīdah wa-Sharī'ah	Maḥmūd Shaltūt	Modern (d. 1963 CE)
Hanafi	al-Fatāwā al-Bazzāziyyah	Muḥammad ibn Muḥammad al-Bazzāzī (al-Kardārī)	Late Classical (d. 827 H / 1424 CE)
Hanafi	Al-Fatawa al-Hindiyyah (al-Fatāwā al-'Ālmgīriyyah)	Committee led by Shaykh Nizām al-Dīn al-Burhānpūrī	Late Classical / Mughal (Commissioned in 1664 CE)
Hanafi	Kitāb al-Mabsūṭ	Shams al-A'immaḥ al-Sarakhsī	Classical (d. 483 H / 1090 CE)
Hanafi	Majallat al-Aḥkām al-'Adliyya	Committee of Ottoman Jurists	Ottoman (Promulgated 1869-1876 CE)
Maliki	Al-Muwatta'	Imām Mālik ibn Anas	Early Classical (d. 179 H / 795 CE)
Maliki	Mukhtaṣar al-'Allāmah Khalīl	Khalīl ibn Ishāq al-Jundī	Classical (d. 776 H / 1374 CE)
Maliki	Al-Qawānīn al-Fiqhiyyah	Ibn Juzayy al-Kalbī	Classical (d. 741 H / 1340 CE)
Maliki	Fatāwā al-Burzulī (Jāmi' Masā'il al-Aḥkām)	Abū al-Qāsim ibn Aḥmad al-Balawī al-Qayrawānī (al-Burzulī)	Classical (d. 841 H / 1438 CE)
Maliki	Fatāwā Ibn Rushd	Abū al-Walīd Muḥammad ibn Aḥmad Ibn Rushd (al-Jadd)	Almoravid (d. 520 H / 1126 CE)
Maliki	Kitab al-Kafi fi Fiqh Ahl al-Madina al-Maliki	Abū 'Umar Yūsuf ibn 'Abd Allāh ibn Muḥammad ibn 'Abd al-Barr al-Namarī al-Qurṭubī	Classical (d. 463 H / 1071 CE)
Maliki	Anwār al-Burūq fī Anwā' al-Furūq	Shihāb al-Dīn al-Qarāfī	Mamluk (d. 684 H / 1285 CE)
Maliki	Fatāwā al-Shaykh al-Imām Muḥammad al-Ṭāhir ibn 'Āshūr	Muḥammad al-Ṭāhir ibn 'Āshūr	Modern (d. 1973 CE)
Shafi'i	Al-Umm	Muḥammad ibn Idrīs al-Shāfi'ī	Early Classical (d. 204 H / 820 CE)
Shafi'i	Al-Hawī al-Kabir	al-Māwardī	Classical (d. 450 H / 1058 CE)
Shafi'i	Fatāwā al-Imām al-Ghazālī	Abū Ḥāmid al-Ghazālī	High Classical (d. 505 H / 1111 CE)

Table 11 (continued)

School	Title	Author	Period
Shafi'i	Al-Fatāwā al-Fiqhiyyah al-Kubrā	Ibn Hajar al-Haytamī	Late Mamluk / Early Ottoman (d. 974 H / 1566 CE)
Shafi'i	Fath al-Wahhāb bi-Sharḥ Manhaj al-Ṭullāb	Shaykh al-Islām Zakariyyā al-Anṣārī	Late Mamluk / Early Ottoman (d. 926 H / 1520 CE)
Hanbali	al-Istikhrāj li-Aḥkām al-Kharāj	Ibn Rajab al-Ḥanbalī	Late Classical (Mamluk) (d. 795 H / 1393 CE)
Hanbali	al-Inṣāf fī Ma'rifat al-Rājih min al-Khilāf	'Alā' al-Dīn al-Mardāwī	Late Classical (Mamluk) (d. 885 H / 1480 CE)
Hanbali	al-Mughnī	Ibn Qudāmah al-Maqdisī	Classical (Ayyubid) (d. 620 H / 1223 CE)
Hanbali	Dalīl al-Ṭālib li-Nayl al-Maṭālib	Mar'ī ibn Yūsuf al-Karmī al-Ḥanbalī	Late Classical (Ottoman) (d. 1033 H / 1624 CE)
Hanbali	Mukhtaṣar al-Khiraqī	Abū al-Qāsim 'Umar ibn al-Ḥusayn al-Khiraqī	Early Classical (Abbasid) (d. 334 H / 945 CE)
Hanbali	al-Furū' wa-Ma'āhu Ṭaṣḥīḥ al-Furū'	Shams al-Dīn Muḥammad ibn Muflīḥ al-Ḥanbalī	Late Classical (Mamluk) (d. 763 H / 1362 CE)
Zahiri	al-Iḥkām fī Uṣūl al-Aḥkām	'Alī ibn Aḥmad ibn Sa'īd ibn Ḥazm al-Andalusī	Classical (d. 456 H / 1064 CE)
Zahiri	al-Muḥallā bi-al-Āthār	'Alī ibn Aḥmad ibn Sa'īd ibn Ḥazm al-Andalusī	Classical (d. 456 H / 1064 CE)
Zaydi	Sharḥ al-Tajrīd fī Fiqh al-Zaydiyyah	Aḥmad ibn Yaḥyā ibn al-Murtaḍā	Classical (d. 840 H / 1437 CE)
Zaydi	Al-Azhār fī Fiqh al-A'imma al-Aṭhār	al-Imām al-Mahdī li-Dīn Allāh Aḥmad ibn Yaḥyā al-Murtaḍā	Late Classical (d. 840 H / 1437 CE)
Zaydi	Sharḥ al-Tajrīd fī Fiqh al-Zaydiyyah	Aḥmad ibn al-Ḥusayn al-Hārūnī al-Ḥasanī	Post-Classical (d. 1055 H / 1645 CE)
Twelver Shi'i (Ja'fari)	Al-Kāfi	Muḥammad ibn Ya'qūb al-Kulaynī	Early Classical (d. 329 H / 941 CE)
Twelver Shi'i (Ja'fari)	Tadhkirat al-Fuqahā'	al-'Allāmah al-Hillī	Late Classical (d. 726 H / 1325 CE)
Twelver Shi'i (Ja'fari)	Tahrīr al-Wasīlah	Ruhollah Khomeini	Contemporary (d. 1989 CE)
Twelver Shi'i (Ja'fari)	Jawāhir al-Kalām fī Sharḥ Sharā'ī' al-Islām	Sheikh Muhammad Hassan al-Najafi	Modern (19th Century) (d. 1266 H / 1850 CE)

B Benchmarking tasks

This section provides detailed descriptions of the task types used to evaluate LLMs' capabilities in Islamic legal reasoning. Our benchmarking framework systematically tests LLMs across three complexity levels, each targeting specific cognitive abilities required for Islamic jurisprudential analysis. The tasks are designed to assess whether LLMs can accurately retrieve information, apply legal rules, and engage in sophisticated reasoning comparable to trained Islamic legal scholars.

Our evaluation framework incorporates both *standard tasks* and *false premise variants*. The false premise variants (marked with prime notation: T1', T2', T3', T7') deliberately introduce incorrect assumptions to test whether models can reject false premise queries (FPQs) instead of confidently fabricating answers. These FPQs are designed in a way to test whether pre-trained models use their internal knowledge to rebut these falsely plausible questions and correct the wrongly stated question (Hu et al. 2023).

For all tasks requiring contextual background (excluding metadata recall tasks T1-T4), we employ a standardized template format that provides comprehensive contextual information before presenting the query. This approach ensures that LLMs have access to all necessary bibliographical and source details without requiring individual manipulation of each query for their respective texts.

Template Format: Context Provision Before Query

```
<Context>:
Book Title: [Title of the Islamic legal text]
Author: [Full name of the author]
Volume-Page Reference: [Specific volume and page numbers where applicable]
Historical Period: [Time period/century of composition or Year of Death of Author]
School of Jurisprudence: [Relevant madhhab where applicable]
Note: [Note for specific queries where applicable]
</Context>
—
Query: [The specific question or task to be performed]
```

This template format is utilized across all complexity levels for tasks that require relevant context for textual analysis, rule extraction, comparative reasoning, and jurisprudential synthesis.

B.1 Evaluation tasks

B.1.1 Low complexity tasks

Low complexity tasks evaluate fundamental knowledge retrieval and basic verification capabilities. These tasks test the model's grounding in Islamic legal scholarship and its ability to access bibliographical and textual information accurately.

T1-T3: Metadata recall These tasks evaluate the LLM's ability to retrieve basic bibliographical and contextual information about Islamic legal texts. LLMs must correctly identify the **jurisprudential school** (*madhhab* ($n = 34$)), **author** ($n = 33$), and **historical period** ($n = 34$) associated with specific works. This tests the model's grounding in the scholarly tradition and grasp of knowledge about Islamic Legal authoritative texts and its ability to distinguish between different schools of Islamic jurisprudence.

Metadata Recall (T1-T3) Examples

T1 - School Identification: What school of thought is associated with book: “Al-Fatawa al-Hindiyyah (also known as al-Fatāwā al-‘Ālamgiriyyah)”? Just return the school of thought without any other text or explanation
Gold Answer: Hanafi

T2 - Author Identification: Who is the author of book: “al-Islām: ‘Aqīdah wa-Sharī‘ah”? Just return the author without any other text or explanation.
Gold Answer: Maḥmūd Shaltūt

T3 - Period Identification: What historical period is book: “Kitāb al-Mabsūṭ” from? Just return the *century* without any other text or explanation.
Gold Answer: 11th century CE (483 H / 1090 CE)

For T3, we initially recorded year-level dates (e.g., 483 AH / 1090 CE) based on available metadata. However, because many of these texts are premodern, sources often disagree on what constitutes the “exact” year—for example, whether to use an estimated publication date, the date of composition, or the author’s death. To eliminate ambiguity and ensure consistency, we standardized the temporal resolution to the century level and report only the Common Era (CE) dates (e.g., 483 H / 1090 CE → 11th century CE).

T4: Concept existence verification ($n = 267$) This task evaluates whether LLMs can determine, with source-specific precision, whether a specified Islamic legal concept is treated in a designated authoritative text. When the concept is present, success requires an accurate binary response of YES or NO.

Concept Existence (T4) Example

Query: Does book: “Mukhtasar al-Quduri” by “Abū al-Ḥasan Aḥmad ibn Muḥammad al-Qudūrī” treat topic: “Khiyar (Options) - Khiyar al-‘Ayb (Option due to defect) in Bay’ (Sales)”? Yes or No? Just return the answer without any other text or explanation.

Gold Answer: Yes, 81

T5: Source of authority or recall ($n = 83$) This task assesses whether a model can identify, within a specified work, the author’s citation of other jurists’ opinions and accurately summarize the cited positions. It tests source-aware attribution, the delineation of intra-school disagreement, and the faithful recall of conditions or qualifications attached to the cited views.

Source Authority (T5) Example

Query: Does the author Muḥammad ibn al-Ḥasan al-Shaybānī in the book “Kitāb al-Buyū” mention any other jurist’s opinion regarding the permissibility of manufacturing contracts (*istiṣnā’*)? Answer Yes/No, and if yes, who?

Gold Answer: Yes. The author mentions the differing opinions of Abū Ḥanīfah (who allows it under specific conditions like known type and weight if an advance is paid) versus Abū Yūsuf and Muḥammad (who allow it more broadly, giving the buyer an option upon seeing the item). (*Qaala Abu Yusuf wa Muhammad: huwa jaa ‘iz wa saahibuhu bakhiiyaar idhaa ra’ahu, in shaa’ akhadhahu wa in shaa’ tarakahu wa laa yakoon ‘abbazlat al-silm*).

T6: Legal Maxim Recall ($n = 13$) This task requires models to extract and succinctly restate specific legal rules or maxims from Islamic legal codes. It evaluates the ability to distill complex principles into precise, self-contained statements while preserving their scope, conditions, and legal effect.

Legal Maxim Recall (T6) Example

Query: According to Article 74 on page 14, what is the legal status of a sale if the seller disposes of the sold item to another person before the first buyer has taken possession?

Gold Answer: If the seller disposes of the sold item before the buyer takes possession, the first sale is rendered void.

B.1.2 Moderate complexity tasks

Moderate-complexity tasks require precise enumeration of legal conditions, comparative analysis, and synthesis of specific rules. They test exactness, concision, and the avoidance of extraneous material while extracting technical requirements from the sources.

T7: Condition enumeration (*Shurūṭ*) ($n = 28$) This task evaluates the LLM's ability to identify and list the specific conditions (*shurūṭ*) required for the validity of Islamic legal transactions. *Shurūṭ* (singular: *shart*) refers to the necessary conditions that must be met for a legal act to be considered valid in Islamic law.

Condition Enumeration (T7) Example

Query: List the specific conditions (*shurūṭ*) for the validity of a *Salām* contract (forward contract) according to the text, as held by Abū Ḥanīfa.

Gold Answer: According to Abū Ḥanīfa, the validity of a *Salām* contract requires the explicit mention of the following conditions in the contract: 1. Known type (*Jins ma'loom*), 2. Known variety (*Naw ma'loom*), 3. Known description (*Wasfah ma'loomah*), 4. Known quantity (*Miqdaar ma'loom*), 5. Known due date (*Ajal ma'loom*), and 6. Knowledge of the capital (*Miqdaar ra's al-maal*). Additionally, the capital must be taken before the parties separate, and neither the capital nor the subject of the contract can be disposed of before possession.

T8: Comparative legal distinction ($n = 21$) This task evaluates the LLM's ability to delineate doctrinal differences between closely related legal rules or applications within Islamic jurisprudence, articulating the operative reasoning that distinguishes similar but legally distinct scenarios.

Comparative Distinction (T8) Example

Query: On page 582, a man guarantees to present a debtor in court the next day, agreeing to pay the one hundred dinar debt if he fails. If he fails to present the debtor, what is the ruling according to Abū Ḥanīfa and Abū Yūsuf, and what is the implicit condition in Muḥammad's differing view?

Gold Answer: According to Abū Ḥanīfa and Abū Yūsuf, the guarantor is liable for the one hundred dinars. Muḥammad's differing view is conditioned on whether the debt was legally proven before the guarantee was made; if it was not, then the claimant's subsequent claim against the guarantor is not considered.

T9: Statute-like synthesis from legal codes ($n = 57$) This task evaluates the LLM's ability to synthesize and interpret codified Islamic legal rules, particularly from systematic legal codes, like the Ottoman *Majalla*. The task requires understanding how specific legal principles apply to practical scenarios.

Statute Synthesis (T9) Example

Query: Based on the discussion on pages 170-171, outline the process and objective of appointing two arbiters (hakamayn) in a case of marital discord (shiqāq).

Gold Answer: In a case of marital discord, the Qur'ān mandates the appointment of two arbiters: one from the husband's family and one from the wife's. The objective of this arbitration is reconciliation (islāh). The arbiters are tasked with investigating the causes of the dispute and attempting to resolve them. If reconciliation is not possible, they are empowered to enact a separation between the spouses. This process is a means of resolving conflict that has escalated beyond the couple's ability to manage.

B.1.3 High complexity tasks

High-complexity tasks probe the model's capacity to identify and articulate underlying jurisprudential reasoning ('*illah*'), compare legal principles across schools, and extend them to novel scenarios. They require methodological fidelity, careful qualification, and rigorous reasoning.

T10: Legal rationale identification ('*illah*') ($n = 63$) This task evaluates the LLM's ability to identify and articulate the underlying legal reasoning ('*illah*') behind specific Islamic legal rulings. '*Illah*' refers to the effective cause or wisdom behind a legal ruling that serves as the basis for analogical reasoning in Islamic jurisprudence.

Legal Rationale Identification ('*illah*') (T10) Example

Query: The text discusses the permissibility of selling a debt (Deen) for a debt (Deen) and the conditions under which it is allowed or prohibited. What is the underlying rationale for the general prohibition of selling a debt for a debt, and what specific condition, if met, allows for its permissibility according to the Hanafi school as presented?

Gold Answer: The underlying rationale for the general prohibition of selling a debt for a debt is to prevent *riba al-nasi'ah* (usury of delay) and *gharar* (uncertainty/risk). When a debt is exchanged for another debt, there is a risk of one party not being able to collect, or the value of the debts changing over time, leading to exploitation or dispute. However, the text implies permissibility if the debt is exchanged for a tangible, present item ('*Ayn haadirah*') that is immediately taken into possession. For example, if a person sells a debt they are owed for a specific, present commodity that they immediately receive, this removes the element of *riba al-nasi'ah* and reduces *gharar*, thus making the transaction permissible. The key condition for permissibility is the immediate and complete possession of at least one of the counter-values in the exchange.

T11: Analogical application to novel scenarios ($n = 12$) This task tests the LLM's ability to apply established Islamic legal principles to contemporary or novel situations not explicitly addressed in classical texts. The task requires understanding of analogical reasoning (*qiyās*) and the ability to identify relevant similarities between classical and modern scenarios.

Analogical Application (T11) Example

Query: Apply the text's principle regarding the prohibition of selling "what is not in one's possession" (Ma laysa fi milkih) to a novel scenario: A person sells a specific car that they intend to buy from a dealership next week, before they have actually purchased it. Detail: 1. Principle, 2. Ambiguity, 3. Ruling based on principle.

Gold Answer:

Principle: The text explicitly prohibits the sale of "what is not in one's possession" (mā laysa fī milkih) at the time of the contract. This principle is rooted in preventing gharar (excessive uncertainty) and ensuring that the seller has the ability to deliver the sold item, thereby avoiding disputes.

Ambiguity: In the novel scenario, the seller has identified a specific car and intends to acquire it, but they do not yet own it at the moment of sale. While the car is identifiable, its ownership and thus the seller's ability to deliver it are contingent on a future event (the purchase from the dealership). This introduces uncertainty regarding the seller's immediate capacity to fulfill the contract.

Ruling based on principle: Based on the principle of prohibiting the sale of what is not in one's possession, this transaction would be considered invalid. Even if the seller fully intends to acquire the car, the lack of actual ownership at the time of the contract makes the sale speculative and falls under the category of gharar. The sale would only become valid once the seller takes possession of the car and then enters into a new contract with the buyer.

T12: Cross-school jurisprudential synthesis ($n = 5$) This task requires LLMs to compare and synthesize different legal positions across multiple schools of Islamic jurisprudence (*madhāhib*). The task tests understanding of inter-school differences and the ability to identify the most widely accepted position (*mashhūr*).

Cross-School Synthesis (T12) Example

Query: Analyze the legal debate on page 353 concerning the effect of a voidable condition (sharṭ fāsīd) on a sales contract. What are the two primary legal opinions on whether such a condition invalidates the entire contract, and what is the jurisprudential basis for each position?

Gold Answer: The text presents two main opinions on the effect of a voidable condition: The condition is void, but the contract is valid: This is the sound position of the Hanbali school. The reasoning is that the sale contract and the condition are two separate legal commitments. The invalidity of the ancillary condition does not nullify the primary contract, which meets all its essential requirements. Both the condition and the contract are void: This view is based on the Prophetic tradition prohibiting "a sale and a condition" (nahā 'an bay' wa sharṭ). Proponents of this view argue that the invalid condition corrupts the entire transaction, rendering the contract itself invalid.

T13: Legal maxim mapping (*Qawā'id Fiqhiyyah*) ($n = 17$) This task evaluates the LLM's ability to identify and apply broad legal maxims (*qawā'id fiqhiyyah*) to specific legal scenarios. *Qawā'id fiqhiyyah* are general principles that encapsulate fundamental concepts in Islamic jurisprudence and serve as interpretive tools for understanding specific rulings.

Legal Maxim Mapping (T13) Example

Query: Article 3 on page 14 states, "In contracts, consideration is given to the intentions and meanings, not to the words and forms." Explain the legal reasoning behind this principle and how it applies to the example of a "sale with right of redemption" (bay' bi-al-wafā').

Gold Answer: The legal reasoning is that the true legal effect of a contract is determined by the parties' actual intent, not merely the literal words they use. In the case of a "sale with right of redemption" (bay' bi-al-wafā'), although the parties use the language of a sale, their intention is to create a security arrangement. Therefore, the law treats it according to its underlying meaning as a pledge (rahn), not as a true sale."

B.2 False premise variants

Just like the "*Wrong Conclusion from False Premises*" error mentioned in Section 2.1, when LLMs accept factually incorrect statements and do not correct the claim, it

can have consequences in Islamic law. A human scholar or *'aalim*, when faced with a query based on a false premise, would first correct the premise. An AI system, if not explicitly designed for this, may simply generate a plausible but entirely incorrect response. Therefore, a simple accuracy score is insufficient for a benchmark in this domain; a robust evaluation must also test a model's ability to identify and correct flawed premises, and to provide sound, verifiable justifications for its conclusions.

B.2.1 Low complexity false premise tasks

T1'-T3': False premise detection (Metadata) These tasks assess a model's ability to detect incorrect attributions by deliberately introducing bibliographical errors. They test whether the model can verify and reject misinformation about Islamic legal texts and scholars. Each variant—false school, false author, and false period contains 15 examples.

False Premise Tasks: Metadata Detection (T1'-T3') Examples

T1' - False School: Analyzing the major Maliki work 'Badā'i' al-Ṣanā'i' fī Tartīb al-Sharā'i', what distinctive systematization methods does it employ that exemplify Maliki approaches to organizing legal principles?

Reason: This question contains a false premise. 'Badā'i' al-Ṣanā'i' fī Tartīb al-Sharā'i' is not particularly influential in Maliki circles - it is actually a major Hanafi legal text.

T2' - False Author: How did al-'Allāmah al-Ḥillī organize the hadith material in his work 'Al-Kāfī'?

Reason: This question contains a false premise. 'Al-Kāfī' was not organized by al-'Allāmah al-Ḥillī. It was actually compiled by Muḥammad ibn Ya'qūb al-Kulaynī.

T3' - False Period: As 'Alā' al-Dīn al-Kāṣānī wrote 'Kitāb al-Mabsūṭ' in the 20th-century CE Modern period, how does it address contemporary Islamic legal challenges?

Reason: This question contains two false premises. First, 'Kitāb al-Mabsūṭ' was not written by 'Alā' al-Dīn al-Kāṣānī but by Shams al-A'immah al-Sarakhsi. Second, it was not written in the 20th-century CE Modern period but is from the Classical period (d. 483 H / 1090 CE), written over 800 years earlier.

B.2.2 Moderate complexity false premise tasks

T7': False premise detection (Condition Enumeration) ($n = 6$) Example This task evaluates LLMs by deliberately attributing legal conditions to incorrect scholars. The task tests whether models can identify when detailed legal conditions are incorrectly attributed to the wrong jurist or school of thought.

False Premise Task: Complex Attribution Detection (T7') Example

Query: According to the ruling on page 145, a marriage contract is formed by offer (ījāb) and acceptance (qabūl) expressed in terms that signify marriage, and it is not valid unless it is witnessed by four free, adult, sane, and Muslim men. What happens if only three witnesses are present?

Reason: This is a false premise question. A marriage contract is formed by offer (ījāb) and acceptance (qabūl) expressed in terms that signify marriage. It is not valid unless it is witnessed by *two* free, adult, sane, and Muslim men, or by one such man and two women, not four men as stated in the question. The false premise incorrectly doubles the witness requirement from two to four men.

C Inter-rate reliability

To assess the consistency between the LLM-based judge and human annotators, we computed Cohen's Kappa scores across two evaluation dimensions: (i) overall scores and (ii) hallucination detection. Table 12 summarizes the inter-rater reliability.

Table 12 Inter-rater reliability between LLM and human annotators using Cohen's Kappa

Comparison	Scores Kappa	Hallucinations Kappa
LLM vs. Human_1	0.7787	0.7671
LLM vs. Human_2	0.8393	0.8650
Human_1 vs. Human_2	0.7320	0.7214
Overall LLM–Human Mean	0.809	0.816

ity between the LLM and each human annotator, as well as the agreement between the two human annotators. The results demonstrate substantial to almost-perfect agreement across all comparisons, with an aggregated LLM–Human reliability of $\kappa = 0.81$.

Author Contributions E.E. created the core benchmarking dataset from 60 foundational texts spanning 1,200 years of Islamic jurisprudence, consisting of 271 data entries. A.M. structured the core dataset, extended it through metadata utilization and automated scripting to 718 entries, developed the complete benchmarking pipeline, including dataset management, evaluation protocols, prompting techniques, experimentation framework, statistical analysis, inter-rater reliability tests, and generated figures, tables, and graphs. A.M. and R.N. wrote the initial drafts of the paper, consisting of: Introduction, Related Work, Dataset Creation, Evaluation Methodology, Setup, Results, Limitations, Discussion, Conclusion, and Appendix sections. J.Q. and I.G. supervised the research, provided guidance on AI and computing methodologies, contributed to study design and experimental framework, and reviewed and refined manuscript drafts. W.I. supervised the research with a focus on NLP and AI approaches, ensured methodological rigor and alignment with state-of-the-art practices, contributed to study design, and provided critical feedback on manuscript drafts. All authors reviewed and approved the final manuscript.

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Data Availability The IslamicLegalBench development set (20% of the complete benchmark) will be made publicly available upon publication at GitHub or HuggingFace, including evaluation instances, task definitions, prompts, and comprehensive documentation. The remaining 80% is maintained as a held-out test set to preserve evaluation integrity and prevent data contamination. Researchers may request an evaluation of the complete benchmark by contacting the authors.

Declarations

Ethical standard This study did not involve human participants as research subjects, animal subjects, or plant subjects. Data annotation was performed by members of the research team as part of the benchmark development process. Accordingly, institutional ethics approval and informed consent were not required.

Competing interests The authors declare no competing interests.

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